



**UNIVERSIDAD
DE GRANADA**

BACHELOR'S THESIS
COMPUTER ENGINEERING

Compensating for Glycemic Ranges Heterogeneity in Glucose Prediction Models via Data Balancing

Autor

Diego Santiago Ortiz

Supervisors

Oresti Baños Legrán

Claudia Villalonga Palliser



HIGHER TECHNICAL SCHOOL OF COMPUTER SCIENCE AND
TELECOMMUNICATIONS ENGINEERING
COMPUTER ENGINEERING, AUTOMATS AND ROBOTICS
Granada, September, 2026

Compensación de la heterogeneidad de los rangos glucémicos en modelos de predicción de glucosa mediante balanceo de datos

Diego Santiago Ortiz

Palabras clave: diabetes tipo 1, monitorización continua de glucosa, predicción de glucosa, desbalanceo de datos, balanceo de datos, SMOTER, SMOGN.

Resumen

Este Trabajo Fin de Grado estudia el efecto del balanceo de los datos de entrenamiento sobre el rendimiento de modelos de predicción de glucosa para personas con diabetes tipo 1. El análisis se realiza utilizando varios conjuntos de datos de monitorización continua de glucosa con diferentes distribuciones entre los rangos glucémicos clínicamente definidos.

Se evalúan tres métodos de balanceo: remuestreo aleatorio, SMOTER y SMOGN. Cada método se combina con distintas configuraciones de balanceo, mientras que la arquitectura predictiva, el procedimiento de entrenamiento y los conjuntos de validación y prueba permanecen sin modificaciones. Los resultados se analizan mediante métricas de regresión y medidas de precisión clínica, tanto de forma global como por rango glucémico.

El objetivo principal es determinar si una representación más equilibrada de los rangos minoritarios durante el entrenamiento permite mejorar la predicción de eventos glucémicos críticos, especialmente los asociados a la hipoglucemia.

Compensating for Glycemic Range Heterogeneity in Glucose Prediction Models via Data Balancing

Diego Santiago Ortiz

Keywords: Type 1 Diabetes, continuous glucose monitoring, glucose prediction, data imbalance, data balancing, SMOTER, SMOGN.

Abstract

This Bachelor's Thesis investigates the effect of balancing the training data on the performance of glucose prediction models for people with Type 1 Diabetes. The analysis is conducted using several continuous glucose monitoring datasets with different distributions across clinically defined glycemic ranges.

Three balancing methods are evaluated: random resampling, SMOTER, and SMOGN. Each method is combined with different balancing configurations, while the predictive architecture, training procedure, validation sets, and test sets remain unchanged. The results are analysed using regression metrics and clinical accuracy measures, both globally and separately for each glycemic range.

The main objective is to determine whether a more balanced representation of minority ranges during training can improve the prediction of critical glycemic events, particularly those associated with hypoglycemia.

Yo, **Diego Santiago Ortiz**, alumno del Grado en Ingeniería Informática de la **Escuela Técnica Superior de Ingenierías Informática y de Telecomunicación de la Universidad de Granada**, con DNI **20889444Q**, autorizo la inclusión de una copia de este Trabajo Fin de Grado en la biblioteca del centro para que pueda ser consultada por las personas interesadas.

Fdo.: Diego Santiago Ortiz

Granada, a [día] de [mes] de 2026.

D. **Oresti Baños Legrán**, profesor del Área de **[Área de conocimiento]** del Departamento de **[Departamento]** de la Universidad de Granada.

Dña. **Claudia Villalonga Palliser**, profesora del Área de **[Área de conocimiento]** del Departamento de **[Departamento]** de la Universidad de Granada.

Informan:

Que el presente trabajo, titulado *Compensating for Glycemic Range Heterogeneity in Glucose Prediction Models via Data Balancing*, ha sido realizado bajo su supervisión por **Diego Santiago Ortiz**, y autorizan su defensa ante el tribunal correspondiente.

Y para que conste, expiden y firman el presente informe en Granada, a **[día]** de **[mes]** de 2026.

Los supervisores:

Oresti Baños Legrán

Claudia Villalonga Palliser

Agradecimientos

Agradecimientos en español

Contents

Acknowledgements	IX
1. Introduction	1
1.1. Context and Background	1
1.1.1. Clinical Burden of Type 1 Diabetes	1
1.1.2. Technological Evolution: From BGM to Continuous Glucose Monitoring	1
1.1.3. Clinical Targets and the Shift towards Time in Range	2
1.1.4. From Glucose Monitoring to Glucose Prediction	2
1.2. Problem Statement	3
1.3. Objectives	3
2. State of the Art	5
2.1. Scope and Methodology	5
2.1.1. Main Research Areas	5
2.1.2. Literature Search and Selection	5
2.2. Glycemic Prediction in Type 1 Diabetes	6
2.2.1. Evolution of Predictive Architectures	6
2.2.2. Prediction Horizons and History Length	6
2.2.3. Evaluation Metrics and Clinical Taxonomy	6
2.3. Data Balancing in Machine Learning and Time Series	8
2.3.1. Data Balancing in Classification	8
2.3.2. Balancing Methods for Regression	9
2.4. Data Balancing Applied to Glycemic Prediction	9
2.4.1. Prevalence of Explicit Balancing	10
2.4.2. Previous Oversampling Approaches	10
2.4.3. Synthetic Data Using Generative Models	10
2.4.4. Patient Heterogeneity as a Contextual Factor	10
2.5. Synthesis and Research Gap	11
2.5.1. Summary of the State of the Art	11
2.5.2. Identified Research Gap	11
3. Exploratory Data Analysis	12
3.1. DiaTrend Dataset	12
3.1.1. Dataset Description	12
3.1.2. Patient Contribution Analysis	13
3.1.3. Glucose Descriptive Statistics	14
3.1.4. Glycemic-Range Distribution	15
3.1.5. Patient-Level Glycemic Distribution	17
3.1.6. Summary of Findings	18
3.2. REPLACE-BG Dataset	19

3.2.1. Dataset Description	19
3.2.2. Patient Contribution Analysis	20
3.2.3. Glucose Descriptive Statistics	21
3.2.4. Glycemic-Range Distribution	22
3.2.5. Patient-Level Glycemic Distribution	23
3.2.6. Summary of Findings	25
3.3. T1DiabetesGranada Dataset	25
3.3.1. Dataset Description	26
3.3.2. Patient Contribution Analysis	26
3.3.3. Glucose Descriptive Statistics	28
3.3.4. Glycemic-Range Distribution	29
3.3.5. Patient-Level Glycemic Distribution	30
3.3.6. Summary of Findings	31
3.4. Cross-Dataset Comparison	32
3.5. Summary and Implications for Data Balancing	34
4. Experimental Methodology	36
4.1. Experimental Methodology Proposal	36
4.1.1. Motivation from the Exploratory Data Analysis	36
4.1.2. Scope of the Balancing Process	36
4.1.3. Proposed Hybrid Balancing Strategy	37
4.1.4. Target Distributions	37
4.1.5. Balancing Methods	40
4.1.6. Patient-Aware Balancing	41
4.1.7. Fixed Experimental Configuration	41
4.1.8. Experimental Workflow	42
4.2. Statistical Analysis and Evaluation	42
4.2.1. Primary Clinical Outcome	43
4.2.2. Primary Effect Measure	43
4.2.3. Statistical Hypothesis	44
4.2.4. Statistical Test	44
4.2.5. Secondary Outcomes	45
4.3. Balancing Implementation	46
4.4. Model Training	47
5. Experimental Results and Discussion	48
5.1. Balance Verification	48
5.2. DiaTrend Results	50
5.2.1. Primary Statistical Test	50
5.2.2. Secondary Outcomes	54
5.2.3. DiaTrend Summary	59
5.3. REPLACE-BG Results	59
5.3.1. Primary Statistical Test	59
5.3.2. Secondary Outcomes	60

5.3.3. REPLACE-BG Summary	63
5.4. T1DiabetesGranada Results	65
5.4.1. Primary Statistical Test	65
5.4.2. Secondary Outcomes	66
5.4.3. T1DiabetesGranada Summary	69
5.5. Cross-Dataset Statistical Synthesis	70
5.5.1. Comparison of the Primary Effects	70
5.5.2. Holm-Adjusted Statistical Results	71
5.5.3. Consistency of the Primary Outcome	71
5.6. Cross-Dataset Secondary-Outcomes Synthesis	71
5.6.1. Influence of the Target Distribution	71
5.6.2. Influence of the Balancing Method	71
5.6.3. Hypoglycemic Performance Across Datasets	72
5.6.4. Global Accuracy and Time-in-Range Trade-Off	72
5.6.5. Hyperglycemic Performance Across Datasets	72
5.7. Main Findings	72
5.8. Future Work	75
6. Project Planning and Budget	76
6.1. Project Planning	76
6.1.1. Initial Workload Distribution	76
6.1.2. Initial Temporal Planning	77
6.2. Project Budget	77
6.2.1. Personnel Costs	77
6.2.2. Computational and Additional Costs	77
6.2.3. Total Estimated Cost	78
A. Source Code Repository	79
B. Experimental Results Repository	80
Bibliography	81

1. Introduction

1.1. Context and Background

1.1.1. Clinical Burden of Type 1 Diabetes

Type 1 Diabetes is a chronic, metabolic disorder defined by the near-total – or even absolute – absence of endogenous insulin production due to the autoimmune destruction of pancreatic beta-cells. As a result, in the absence of enough insulin in the system, glucose cannot be used properly for its corresponding primary functions. Unlike Type 2 Diabetes, which primarily involves resistance to insulin and progressive beta-cell dysfunction, T1D typically manifests with a rapid set of symptoms and an immediate risks if untreated. This physiological deficit requires lifelong, intensive exogenous insulin replacement therapy in order to maintain metabolic homeostasis. Considering that every daily decision directly influences the precarious balance of blood glucose levels of a person, the clinical burden placed upon those individuals with T1D is exceptionally high, as they must continuously act as an "external controller" for such complex and highly sensitive regulatory system. The failure to “externally” achieve this balance leads to the following two distinct categories of clinical risk:

- Hypoglycemia (low blood sugar). It is fixed by a correction dose of sugar. It leads to immediate impairment, seizures, loss of consciousness, and death.
- Chronic hyperglycemia (high blood sugar). It is fixed with a correction dose of insulin. It leads to long-term microvascular and macrovascular damage.

1.1.2. Technological Evolution: From BGM to Continuous Glucose Monitoring

The management of T1D has undergone a paradigm shift in recent years, driven by the evolution of monitoring technologies. For decades, the standard of care was limited to Blood Glucose Monitoring (BGM) via intermittent capillary finger-sticks and the measurement of Hemoglobin A1c (HbA1c). While HbA1c provides a reliable indicator of average glucose levels over the preceding 90 days – and it actually still remains a "golden" standard for predicting long-term glycemic risks – it is fundamentally limited in its ability to capture glycemic variability. Patients with identical average HbA1c values can have vastly different glycemic profiles: one may show stable normoglycemia all the time, while another frequently suffers from dangerous fluctuations between extreme high and low values. The introduction of Continuous Glucose Monitoring (CGM) systems has provided the high-resolution time-series data necessary to visualize these kind of fluctuations. These devices

use sub-cutaneous sensors to measure glucose concentrations, with measurements typically recorded every 5 to 15 minutes. In addition, real-time CGM (rtCGM) systems, continuously transmit data to a receiver such as smartphone, allowing for the implementation of predictive alerts and alarms.

1.1.3. Clinical Targets and the Shift towards Time in Range

With the widespread of CGM, it was also necessary to establish more nuanced standard glycemic ranges, instead of relying on simple means. As a result, the International Consensus on Time In Range (TIR) [1] defined metrics that describe the actual reality of T1D clinical management:

- Time Below Range (TBR) -> The percentage of time a patient's glycemic level is below 70 mg/dL. It can be divided into:
 - Level 1 -> 54–69 mg/dL
 - Level 2 -> <54 mg/dL

Minimizing TBR is actually the highest priority for clinical safety, since Level 2 hypoglycemia represents an immediate critical medical emergency, as previously described.
- Time In Range (TIR) -> The percentage of time a patient's glycemic level is between 70 and 180 mg/dL.
- Time Above Range (TAR) -> The percentage of time a patient's glycemic level is above 180 mg/dL. It can also be divided into:
 - Level 1 -> 181–250 mg/dL
 - Level 2 -> >250 mg/dL

This ranges establish the necessary essential consensus for evaluating treatments efficacy and patient's safety.

1.1.4. From Glucose Monitoring to Glucose Prediction

Even to this day, the most common approach in the daily management of T1D remains reactive: a glucose level is detected, and an insulin dose is subsequently adjusted from that only measurement. However, the widespread adoption of Continuous Glucose Monitoring (CGM) is enabling a conceptual paradigm shift: using AI to predict critical glycemic values in advance.

While reactive management relies on correcting "dys-glycemia" after it has already occurred, proactive AI aims to forecast glycemic trends 30 to 60 minutes into the future-seeking in order to act before the problem has already materialized.

This "early warning system" holds immense clinical value: by anticipating these critical up-coming values toward either hypoglycemia or hyperglycemia, with a window of several minutes, patients can intervene earlier,

with less aggressiveness and with less potential risk. In summary, this transition from "detecting" to "predicting" not only improves glucose stability but also significantly reduces the physical and emotional impact on the patient, allowing for a more seamless and less stressful management experience.

1.2. Problem Statement

Despite the high availability of glycemic data thanks to CGM and the rapid development of AI these days, the development of robust glycemic forecasting models is still plagued by the "Lazy AI" phenomenon. This term refers to: *models that achieve high global accuracy by essentially "playing it safe" – optimizing for the most common states while failing to predict the critical, low-frequency events that actually represent the greatest risk to the patient.*

In T1D glucose prediction, this phenomenon occurs due to the inherent imbalance of real-world glycemic data.

In a typical well-controlled – or even moderately-controlled – patient, the vast majority of measurements fall within the normoglycemic (70-180 mg/dL) or Level 1 hyperglycemic (181-250 mg/dL) ranges. In contrast, hypoglycemic events (< 70 mg/dL) often represent 4% or less of the total historical data points of a patient.

Most machine learning algorithms are designed to minimize a global error metric, such as MSE, RMSE, or similar, which implicitly gives equal importance to all the points. This makes the model "play it safe" in the training by assuming that the patient will remain within the normoglycemic range, which actually gets an "excellent" error rate.

This is the essence of Lazy-AI: the model avoids predicting rare values of the spectrum in order to minimize the risk of large error penalties. This mean-seeking behavior, specifically in T1D glucose prediction, creates a "majority bias" that makes it fail to trigger the corresponding necessary life-saving alarms for the most critical values.

1.3. Objectives

The aim of this thesis is to investigate the impact of data balancing techniques on the training and clinical performance of glucose prediction models for T1-Diabetes. It focuses on the study and re-distribution of CGM datasets, where glucose values are naturally imbalanced across glycemic ranges: the majority of samples belong to the normoglycemic range, while hypo and hyper-glycemia are much more under-represented.

To achieve this general objective, the following specific objectives are defined:

- OE0. Review the State of the Art on data balancing for glucose predictions.

First, a preliminary review of the scientific literature is conducted to identify existing data balancing methodologies for glucose prediction using machine learning. If the available literature is found to be limited on this matter, the review will focus separately on: machine learning algorithms for glucose prediction, and data-balancing techniques in related time-series regression or healthcare prediction problems.

- OE1. Analyze the distribution of glycemic ranges in the selected datasets.

This thesis works with three different CGM datasets: DiaTrend, REPLACE-BG and T1DiabetesGranada. They will be thoroughly examined and analyzed to quantify the degree of imbalance in real-world glycemic data, by clearly identifying TIR, TBR and TAR proportions.

- OE2. Apply data balancing techniques to the training data.

Balancing strategies such as SMOTER [2], SMOGN [3], or similar over-sampling techniques will be studied, implemented, and applied to generate alternative training sets with improved representation of previously under-represented glycemic regions.

- OE3. Train existing glucose prediction models under fixed experimental conditions.

The same pre-existing models – inherited by the research team tutoring the thesis – will be trained using the original imbalanced datasets and the different balanced versions, ensuring that any observed differences in performance can then be attributed to the data balancing process carried out.

- OE4. Evaluate the statistical and clinical impact of balancing glucose datasets.

Model performance will be assessed using both conventional regression metrics – RMSE, MSE, etc – and clinically meaningful indicators – such TIR, Clarke Error Grid, etc. This will allow to distinguish between numerical accuracy and clinical impact.

2. State of the Art

2.1. Scope and Methodology

The aim of this chapter is to review the current literature carried out in the field of T1-Diabetes glycemic prediction. Additionally, different data-balancing techniques proposed for time-series prediction are studied in this section

2.1.1. Main Research Areas

In order to build a solid state-of-the-art review, this chapter distinguishes between, at least, the following three overlapping but non-equivalent research areas:

- T1-Diabetes prediction using CGM data
- Data balancing techniques applied to generic machine learning problems and time-series forecasting.
- Data balancing approaches for T1 Diabetes prediction

2.1.2. Literature Search and Selection

To perform this review, several academic repositories and scientific databases were consulted, including *Google Scholar* [4], *PubMed* [5], *IEEE Xplore* [6], *ScienceDirect* [7], and *SpringerLink* [8]. The search process utilized keywords such as:

- *"Type 1 Diabetes"*
- *"CGM forecasting"*
- *"Glycemic prediction"*
- *"Data balancing"*
- *"Imbalanced regression"*
- *"SMOTER"*
- *"SMOBN"*

The selection of literature was then filtered by its relevance to the scope of this study.

2.2. Glycemic Prediction in Type 1 Diabetes

2.2.1. Evolution of Predictive Architectures

Glycemic prediction has recently evolved from simple statistical and physiological approaches towards machine learning. Broad reviews from 2019 [9] and 2022 [10] state how RNN, SVM, Gaussian and hybrid approaches currently co-exist, while also emphasizing the fact that, for now, there is not an universal model that performs excellently for all clinical contexts.

Recent works [9, 10] follow-up with: LSTM, GRU-based, proposals incorporate attention mechanisms, Transformers and GNN to capture higher complexity.

Furthermore, literature reviews [10] state the field has bifurcated into two families: continuous regression of future glucose values and event prediction (primarily hypoglycemia), formulated as a classification problem for early warnings.

2.2.2. Prediction Horizons and History Length

The scientific literature [9, 10, 11, 12] converges on the idea that short-medium horizons – 30 min to 1 hour – are the most realistic and clinically useful. This range (30 to 60 min horizon) appears consistently in reviews of the T1D prediction literature. Also, evidence shows how the error increases rapidly as this horizon lengthens.

Regarding the length of history window, literature also converges on the idea that most reusable evidence for decision support is concentrated in windows of at least up to one hour before [9, 10, 11, 12].

2.2.3. Evaluation Metrics and Clinical Taxonomy

The evaluation of glycemic prediction requires both statistical and clinical metrics.

Most studies [9] in the field use the typical regression problems measures: RMSE, MAE, MSE, etc; since they provide the most direct and uncomplicated numerical estimation of the difference between the predicted and the real values, and allow comparing models in terms of "average numerical accuracy".

However, for problems like this, average error metrics may not be sufficient: a model can obtain a low RMSE while still failing to detect rare but clinically dangerous events. Therefore, these models' evaluation must also consider where the error occurs, not only how large the average error is. For this reason, glucose prediction studies always include additional clinically-oriented metrics. As described in 1.1.3, the standard consensus amongst all literature is "glycemic ranges", often referred as **Time In Range** [1]. These metrics are specially relevant for this thesis because they clearly define separate ranges according to their actual clinical relevance, which will allow us to directly quantify the imbalance of the glucose distribution.

Another widely used clinical evaluation tool is the Clarke Error Grid Analysis (CEGA) [13]. Unlike other generic regression metrics, CEGA not only reflects the numerical difference between the predicted and the real values, but it also classifies those predictions into 5 different categories according to their potential clinical consequences. From safe to critical:

- **Zone A.** Clinically accurate.
- **Zone B.** Slight deviation from ground truth but still clinically safe.
- **Zone C.** Errors that could lead to unnecessary correction.
- **Zone D.** Dangerous failures to detect hypo- or hyperglycemia.
- **Zone E.** Critical errors that lead to the opposite.

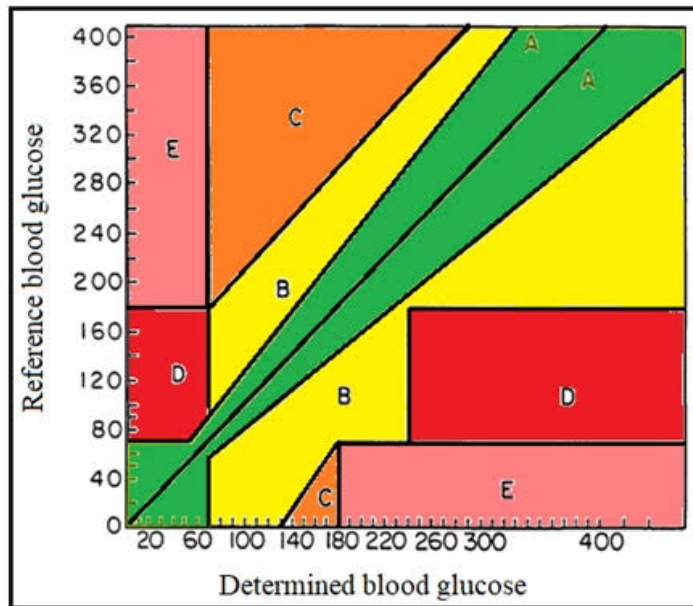


Figure 2.1: Clarke Error Grid, classifies glucose predictions according to their clinical risk.

This taxonomy is particularly important when considering clinical impact: a prediction error in the normo-glycemic may be harmless for the patient, while a similar absolute error near the critical ranges may dangerously prevent an alarm from being triggered. Therefore, apart from considering numerical regressions errors, a proper model evaluation must also consider this kind of clinical metrics.

Inspired by the Clarke Error Grid, recent research has also studied the possibility of integrating clinical awareness into the learning process itself,

through the implementation of *glycemic-aware loss functions*. For example, *glucose-specific MSE* [14] extends the traditional MSE by applying higher penalties to clinically harmful errors. Similarly *Hypo-Hyper Loss* [15] introduces asymmetric penalties depending on whether the prediction error affects hypoglycemic or hyperglycemic regions.

2.3. Data Balancing in Machine Learning and Time Series

2.3.1. Data Balancing in Classification

The problem of data imbalance was first addressed in classification problems. Canonical techniques include the following ones:

SMOTE

Synthetic Minority Oversampling Technique [16], is an oversampling method used to address class imbalance in classification problems, by generating artificial minority-class instances in the feature space. For each selected minority sample, SMOTE identifies its k -nearest neighbors within the same minority class and creates a new synthetic sample by linear interpolation between the original point and one randomly chosen neighbor. The new instance is randomly generated along the segment connecting both samples. While naive duplication of samples would only increase density inside the minority class, this approach introduces more diversity to the group, helping the classifier learn a less biased decision boundary toward the majority class.

Borderline-SMOTE

Borderline-SMOTE [17] is an extension of SMOTE that also generates synthetic samples, but only around the minority-class instances that are located near the decision boundary, where misclassification is more likely. Synthetic examples are then created by interpolating between these "borderline" instances and their minority-class neighbors. This approach makes the oversampling process focus on the most informative and vulnerable regions of the feature space, helping the classifier perform better near the boundaries.

ADASYN

Adaptive Synthetic Sampling [18] is an oversampling technique that addresses data balancing by focusing on areas where learning is harder. Like SMOTE, ADASYN generates synthetic samples via interpolation between minority-class near neighbors. However, ADASYN "adaptively" assigns a higher number of synthetic samples to those minority instances that are

surrounded by more majority-class neighbors, since these points are the most difficult to classify.

2.3.2. Balancing Methods for Regression

In regression, the situation is more complex because discrete boundaries do not exist. There is no such thing as "rare regions" because all the points are inherently considered to be equally important. This differentiation is key, and directly answers why data-balancing in regression has historically been treated much less than in classification. However, some extensions of the previous techniques have arisen from this same concern:

SMOTE-R

SMOTE for Regression [2], is an extension of SMOTE for imbalanced regression problems, where the target variable is continuous rather than categorical. Instead of oversampling a minority class, SMOTE-R focuses on rare or highly relevant target-value "regions" – such as extreme glucose values in this case. Like SMOTE, it generates synthetic samples by interpolating between near neighbors, but determines which areas should be oversampled according to their relevance of the continuous target.

SMOEN

Synthetic Minority Oversampling Technique for Regression with Gaussian Noise [3], is an extension of SMOTE-R that generates synthetic samples in highly relevant target regions by combining simple interpolation with the addition of Gaussian noise. This makes the generated data more diverse and helps to avoid the limitations of simple interpolation in cases where rare regions are irregularly distributed.

Recent works has arisen other – while much less adopted in the literature – data balancing techniques such as: **SERA** [19], which evaluates regression error by giving greater importance to rare or highly relevant target values; **Balanced MSE** [20], which extends MSE to reduce of highly-frequent target values during training; and **Geometric variants** [21], which redefine the synthetic samples generation to consider the geometric distribution of the local area instead of simple interpolation.

2.4. Data Balancing Applied to Glycemic Prediction

The importance of this job does not lie on studying glycemic prediction and data-balancing techniques separately, but on identifying what have been done in combination of both, to improve model behavior in rare but clinically critical glycemic regions.

2.4.1. Prevalence of Explicit Balancing

When narrowing the scientific research to the combined focus of this thesis, explicit balancing appears much less than expected in glycemic prediction studies. Literature reviews [10] can mostly be categorized into two big areas: glucose prediction and event prediction; but balancing the continuous target is hardly ever considered the central theme.

2.4.2. Previous Oversampling Approaches

Mayo et. al [22] author one of the only – if not the only – studies that uses “glycemic-aware” metrics and oversampling techniques for predicting blood glucose. They demonstrate two important things: first, that the best model highly depends on the glycemic sub-range, there is not a model that performs well for all; and second, that oversampling helps improving prediction performance specifically in hypoglycemia.

Mayo et al. use Random Oversampling, SMOTE, and ADASYN, to generate synthetic samples for the minority ranges. In their results, an MLP (Multi-Layer Perceptron) trained with the oversampled data was the best option for the hypoglycemic range, while a linear SVR (Support Vector Regression) performed better in the normal and hyperglycemic ranges.

2.4.3. Synthetic Data Using Generative Models

In this other work by Nogher et. al [23], they explore the possibility of using generative deep learning-based architectures to generate synthetic data. This idea arises from the conception that traditional mathematical generators often fail to accurately replicate the complex, individualized physiological behaviors of real patients.

In their results, they demonstrate that training models with datasets that have been synthetically oversampled with the mentioned mechanisms, achieves a general improvement across all evaluated metrics. This work sets the precedent in the use of generative machine learning techniques to train glucose prediction model.

2.4.4. Patient Heterogeneity as a Contextual Factor

While not directly relevant to the scope of this thesis, it is also interesting to consider other related literature that shows that patient characteristics – such as age, gender, or comorbidities – may affect the frequency and predictability of different glycemic ranges. Recent research has explored the role of patient heterogeneity in glucose prediction. Baños, Villalonga et al. [24] studied the impact of factors such as age, gender, hypertension, obesity, and hypothyroidism on the performance of Linear, LSTM, and CNN models using the T1DiabetesGranada dataset. Their results showed that incorporating patient-specific factors can improve prediction accuracy, particularly for elderly patients and those with hypertension or obesity.

2.5. Synthesis and Research Gap

2.5.1. Summary of the State of the Art

The literature review performed in this chapter clearly shows how previous works have extensively studied glucose forecasting, event prediction, clinical evaluation metrics, and different forms of data augmentation or oversampling. However, it also shows that most glucose prediction models are still being evaluated through typical regression global error metrics, which may be hiding important clinical risks.

2.5.2. Identified Research Gap

Although previous studies have separately explored glucose forecasting, clinical metrics, and data-balancing mechanisms, there is still limited evidence on whether balancing the training data according to glycemic ranges proportions can improve model performance in underrepresented but clinically-critical regions.

Therefore, the gap addressed in this thesis can be formulated through the following question: **can data-balancing techniques – such as SMOTER, SMOGN, or related approaches – improve predictive performance across imbalanced glycemic regions without substantially degrading overall accuracy?**. This study investigates this hypothesis and evaluates its validity.

In order to address this question, this thesis will analyze the distribution of glucose values in three CGM datasets (DiaTrend, REPLACE-BG and T1DiabetesGranada), **apply balancing techniques according to clinically defined glycemic ranges**, train existing predictive models under controlled experimental conditions, and compare their performance statistically.

3. Exploratory Data Analysis

This chapter presents the exploratory data analysis performed on the three CGM datasets used in this thesis: DiaTrend, REPLACE-BG, and T1DiabetesGranada. These datasets had already been standardized during preprocessing, so this chapter does not include an extensive technical inspection of column formats or data normalization. The actual purpose of this analysis is to describe the structure, scale, and distribution of the glucose measurements, in order to quantify the degree of imbalance across glycemic ranges.

The chapter is organized in the following way: first, each dataset is analyzed independently using the same steps; then, a cross-dataset comparison is presented to show differences and similarities; and finally, the implications of the EDA for the data balancing methodology are summarized.

3.1. DiaTrend Dataset

3.1.1. Dataset Description

The dataset includes both a patient-level information table that provides demographic information, and a glucose measurement table with the timestamped glucose values.

The patient information table includes the following variables: patient identifier (`patient_id`), sex (`sex`), age (`age`), number of days with measurements (`measurement_days`), and total number of measurements (`measurement_count`).

The glucose measurement table includes the following variables: patient identifier (`patient_id`), measurement date (`date`), measurement time (`time`), glucose value (`glucose`), and temporal alignment columns at 5-minute and/or 15-minute resolution (`timestamp_5min` and `timestamp_15min`).

Table 3.1: General description of the DiaTrend dataset.

Metric	Value
Number of patients	51
Total number of glucose measurements	7,667,356
Start date	2015-12-01 21:01:07
End date	2022-06-28 23:59:34
Total recording span	2,401 days
Number of days with measurements	2,358
Rows with 5-minute alignment	8,750,125
Rows with 15-minute alignment	2,916,712

Additionally, a consistency check was performed between the patient metadata table and the measurements table. Although the metadata file contains information for 54 patients, only 51 patients are present in the

measurement dataset. Specifically, patients 50, 52, and 54 appear in the metadata table but do not have associated glucose measurements in the processed dataset. Therefore, analysis will be based on the 51 patients for whom glucose records are available on the measurements table.

Table 3.2: Patient information summary for the DiaTrend dataset.

Metric	Value
Mean age	30.57
Median age	25
Minimum age	19
Maximum age	69
Male patients	16
Female patients	35
Mean days with measurements per patient	540.37
Median days with measurements per patient	261
Mean measurements per patient	150,340.31
Median measurements per patient	80,488

This generic structural description provides the basic context needed for the following analyses

3.1.2. Patient Contribution Analysis

For each patient, the number of glucose measurements and the number of days with at least one measurement were computed and summarised in Table 3.3

Table 3.3: Patient contribution summary for the DiaTrend dataset.

Metric	Value
Mean measurements per patient	150,340.31
Median measurements per patient	80,488
Minimum measurements per patient	9,075
Maximum measurements per patient	475,992
Mean days per patient	540.37
Median days per patient	261
Minimum days per patient	33
Maximum days per patient	1,907
Percentage contributed by the largest patient	6.21%
Percentage contributed by the top 5 patients	29.32%
Percentage contributed by the top 10 patients	53.27%
Number of patients contributing 50% of the dataset	10

Figure 3.1 shows the number of measurements per patient, sorted in descending order.

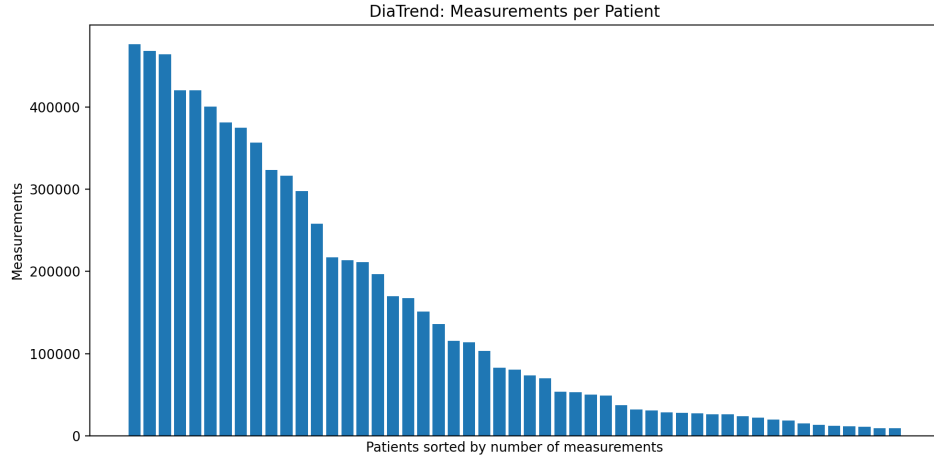


Figure 3.1: Measurements per patient in DiaTrend.

In this dataset, patient contribution to the total glucose measurements distribution is highly uneven, as clearly shown by the descending pattern of Figure 3.1.

Quantitatively, as shown in Table 3.3, the largest patient contributes 6.21% of all measurements, while the top 5 patients account for 29.32% and the top 10 patients account for 53.27% of the dataset. In other words, approximately half of all available glucose measurements come from only 10 out of the 51 analyzed patients, which makes the global distribution highly influenced by the glycemic profiles of those individuals with the larger monitoring history. While this finding does not necessarily introduces a bias towards the scope of this thesis – which is mostly agnostic to individual patient characteristics, it is still relevant enough to be taken into consideration for the subsequent analyses.

3.1.3. Glucose Descriptive Statistics

This section provides a general view of the global distribution of glucose values.

Table 3.4: Glucose descriptive statistics for the DiaTrend dataset.

Metric	Value
Minimum glucose	39 mg/dL
Maximum glucose	401 mg/dL
Mean glucose	186.33 mg/dL
Median glucose	175 mg/dL
Standard deviation	74.18 mg/dL

As shown in Figure 3.2, the glucose distribution follows a typical diabetes-

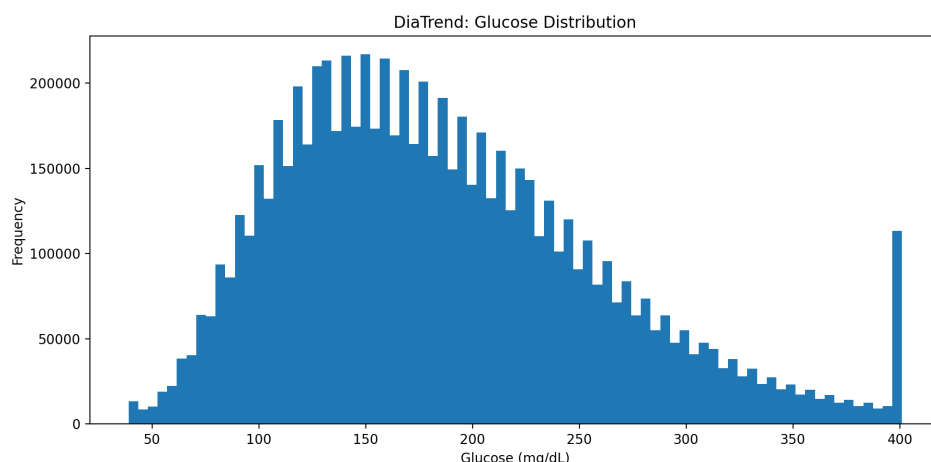


Figure 3.2: Glucose distribution in DiaTrend.

related pattern: the highest concentration of measurements appears around the central normoglycemic and early hyperglycemic regions, while the frequency gradually decreases towards the more extreme hypoglycemic and severe hyperglycemic values.

It is also interesting to note how the histogram presents an irregular pattern with alternating peaks and lower bars. This “one peak, one drop” effect may be related to the measurement resolution, rounding, or preprocessing of CGM values, but it does not change the general shape of the distribution.

3.1.4. Glycemic-Range Distribution

While the previous section describes glucose distribution globally, this one focuses on the analysis of clinically relevant glycemic ranges. As for a reference, the following glycemic ranges are formally presented in Table 3.5:

Table 3.5: Glycemic range definitions.

Glycemic range	Definition
Severe hypoglycemia	< 54 mg/dL
Hypoglycemia	54–69 mg/dL
Normoglycemia	70–180 mg/dL
Hyperglycemia	181–250 mg/dL
Severe hyperglycemia	> 250 mg/dL

The count and the derived percentage of glucose measurements were computed for each of the previous ranges in DiaTrend.

Figure 3.3 makes the imbalance towards clinically critical glycemic ranges clearly visible. Quantitatively, as described in Table 3.6, normo-glycemia is the dominant range representing 50.88% of all glucose values, hyperglycemia

Table 3.6: Glycemic range distribution in DiaTrend.

Glycemic range or category	Count	Percentage
Severe hypoglycemia	35,291	0.46%
Hypoglycemia	105,621	1.38%
Normoglycemia	3,901,129	50.88%
Hyperglycemia	2,192,705	28.60%
Severe hyperglycemia	1,432,610	18.68%
TBR (< 70 mg/dL)	140,912	1.84%
TIR ($70\text{--}180$ mg/dL)	3,901,129	50.88%
TAR (> 180 mg/dL)	3,625,315	47.28%

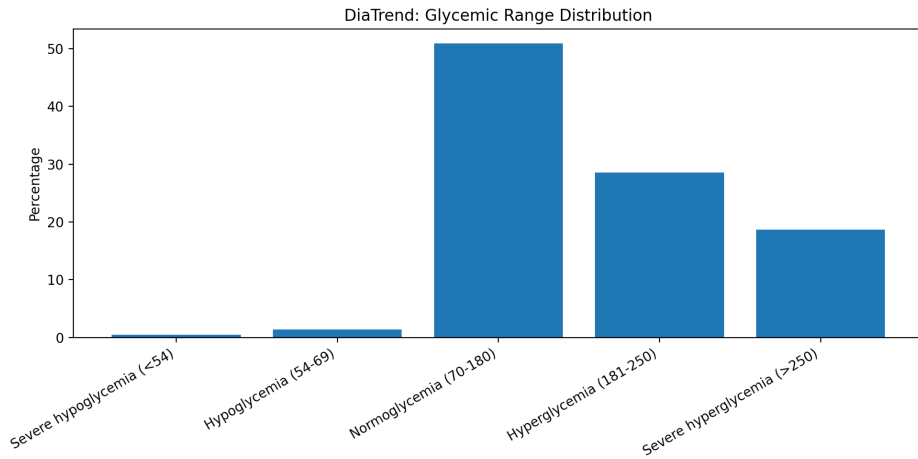


Figure 3.3: Glycemic range distribution

and severe hyperglycemia are also highly represented accounting together for the 47.28% of the measurements, while hypoglycemia and severe hypoglycemia only account together for the 1.84% of the total

Table 3.7: Imbalance ratios in DiaTrend.

Ratio	Value
Normoglycemia / hypoglycemia	27.68
Normoglycemia / severe hypoglycemia	110.54
Normoglycemia / severe hyperglycemia	2.72
Majority range / smallest minority range	110.54
TIR / TBR	27.68
TIR / TAR	1.08

Additionally, the specific imbalance ratios were also quantified in Table 3.7, in order to express the imbalance in more direct quantitative terms, that is: "how much more frequent the majority ranges are compared the clinically critical ones". For example, in DiaTrend, normoglycemic samples appear 27.68 times more frequently than hypoglycemic samples and 110.54 times more frequently than severe hypoglycemic samples.

3.1.5. Patient-Level Glycemic Distribution

While the global glycemic percentages presented in the previous section are useful, they may hide patient-level differences. For this reason, in order to make a more representative analysis of all the patients, the glycemic ranges distribution was also computed independently for each of them. A summarized version of these results is presented in Table 3.8

Table 3.8: Patient-level glycemic range statistics in DiaTrend.

Metric	Value
Mean patient TBR	1.67%
Median patient TBR	1.18%
Minimum patient TBR	0.06%
Maximum patient TBR	9.31%
Mean patient TIR	53.97%
Median patient TIR	52.52%
Minimum patient TIR	16.03%
Maximum patient TIR	79.57%
Mean patient TAR	44.36%
Median patient TAR	46.50%
Minimum patient TAR	18.96%
Maximum patient TAR	83.87%

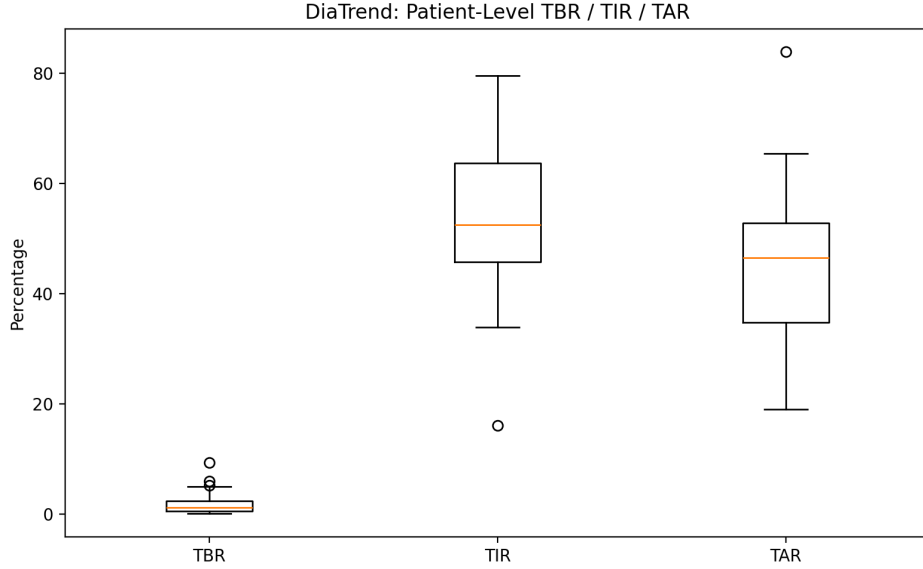


Figure 3.4: Glucose distribution in DiaTrend.

Figure 3.4 makes patient-level distribution much more interpretable by presenting it in the form of a box-plot: each box summarises how each of ranges varies for the different patients. This representation allows to interpret whether the imbalance ratios described in general terms are actually equivalent for each patient of the dataset population.

The TBR boxplot is concentrated very close to zero, with a median of 1.18% and a maximum value of 9.31%. This indicates that hypoglycemic measurements are mostly consistently rare for most patients, not only at global dataset level. On the contrary, TIR and TAR show much wider variability: TIR has a median of 52.52%, but ranges from 16.03% to 79.57%, while TAR has a median of 46.50% and ranges from 18.96% to 83.87%. Therefore, although TIR is the majority range for 48 out of 51 patients, the balance between normoglycemia and hyperglycemia may vary substantially between different patients. In any case, the plot confirms the consistent under-representation of hypoglycemic values across patients, while normo and hyper-glycemia depends more on the patient.

3.1.6. Summary of Findings

DiaTrend contains 51 patients and 7,667,356 valid glucose measurements. The dataset covers the period from 2015-12-01 21:01:07 to 2022-06-28 23:59:34, with a total recording span of 2,401 days.

The patient contribution analysis shows that DiaTrend is highly uneven in terms of per-patient measurements availability. The largest patient contributes 6.21% of all measurements, while the top 10 patients account for

53.27% of the dataset.

The glucose distribution analysis shows a normo-diabetes pattern: the highest concentration of measurements appears around the central normoglycemic and early hyperglycemic regions, while the frequency gradually decreases towards the more extreme hypoglycemic and severe hyperglycemic values.

The imbalance ratios confirm this pattern quantitatively: normoglycemic samples appear 27.68 times more frequently than hypoglycemic samples and 110.54 times more frequently than severe hypoglycemic samples, while hyperglycemia and severe hyperglycemia appear together globally almost as much as normoglycemia.

At patient level, TBR remains consistently low across patients, which confirms that hypoglycemia is not only globally rare but also generally underrepresented across population. On the other hand, TIR and TAR show wider variability, indicating that the balance between normoglycemia and hyperglycemia highly depends on patient’s characteristics.

3.2. REPLACE-BG Dataset

Since the subsequent analyses for REPLACE-BG are analogous to the ones performed for Diatrend, some of the EDA methodology has been omitted.

3.2.1. Dataset Description

Like Diatrend, REPLACE-BG dataset includes both a patient-level information table that provides demographic information, and a glucose measurement table with the timestamped glucose values. Tables’ fields are also analogous to those described for Diatrend 3.1.1. The main descriptive characteristics of the dataset are summarized in Table 3.9

Table 3.9: General characteristics of the REPLACE-BG dataset.

Metric	Value
Number of patients	223
Total number of glucose measurements	10,432,408
Start date	2000-01-13 00:01:53
End date	2000-11-25 14:39:43
Total recording span	317 days
Number of days with measurements	318
Rows with 5-minute alignment	11,528,724
Rows with 15-minute alignment	3,842,908

A consistency check was also performed for REPLACE-BG between the patient metadata table and the measurements table. The metadata file contains information for 223 patients, and all 223 patients are also present in

the measurement dataset. Therefore, since no patients appear in one table but not in the other, all subsequent analyses for REPLACE-BG are based on the full set of 223 patients. Table 3.9 summarises their demographic characteristics.

Table 3.10: Patient information summary for REPLACE-BG.

Metric	Value
Mean age	44.17
Median age	43
Minimum age	19
Maximum age	78
Male patients	111
Female patients	112
Mean days with measurements per patient	172.95
Median days with measurements per patient	178
Mean measurements per patient	46,570.37
Median measurements per patient	48,475

This generic structural description provides the basic context needed for the following analyses.

3.2.2. Patient Contribution Analysis

For each patient, the number of glucose measurements and the number of days with at least one measurement were computed, and summarised in Table 3.11

Table 3.11: Patient contribution summary in REPLACE-BG.

Metric	Value
Mean measurements per patient	46,782.10
Median measurements per patient	48,688
Minimum measurements per patient	10,013
Maximum measurements per patient	54,932
Mean days per patient	172.95
Median days per patient	178
Minimum days per patient	38
Maximum days per patient	200
Percentage contributed by the largest patient	0.53%
Percentage contributed by the top 5 patients	2.56%
Percentage contributed by the top 10 patients	5.08%
Number of patients contributing 50% of the dataset	104

Figure 3.5 shows that REPLACE-BG has a much more homogeneous patient contribution than DiaTrend: bars decrease gradually rather than

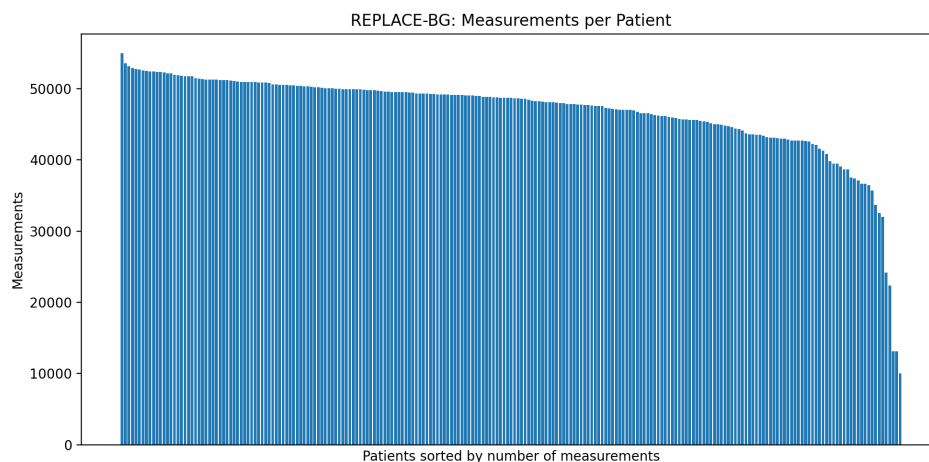


Figure 3.5: Measurements per patient in REPLACE-BG.

showing a sharp drop.

Quantitatively, as shown in Table 3.11, the largest patient contributes only 0.53% of all measurements, the top 10 patients contribute 5.08%, and 104 out of the 223 patients are required to account for 50% of the dataset. Compared with DiaTrend, where the top 10 patients accounted for more than half of all measurements, REPLACE-BG is much less dominated by a small subset of individuals. Therefore, from a patient-contribution perspective, REPLACE-BG is much more balanced.

3.2.3. Glucose Descriptive Statistics

The global distribution of glucose values was also analyzed and summarised in Table 3.12.

Table 3.12: Glucose descriptive statistics in REPLACE-BG.

Metric	Value
Minimum glucose	39 mg/dL
Maximum glucose	401 mg/dL
Mean glucose	159.89 mg/dL
Median glucose	150 mg/dL
Standard deviation	64.13 mg/dL

As shown in Figure 3.6, the glucose distribution follows a typical diabetes-related pattern: the highest concentration of measurements appears around the central normoglycemic and early hyperglycemic regions, while the frequency gradually decreases towards the more extreme hypoglycemic and severe hyperglycemic values.

The histogram also presents an irregular “one peak, one drop” pattern,

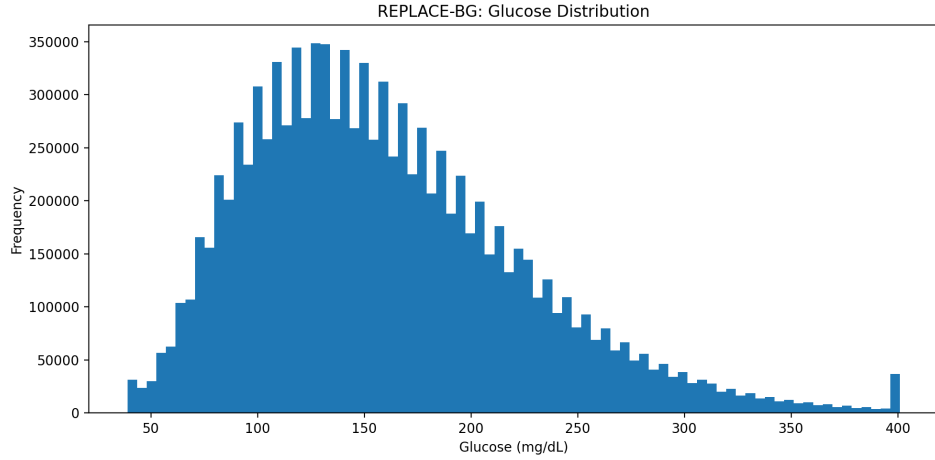


Figure 3.6: Glucose distribution in REPLACE-BG.

that may be related to the measurement resolution. However, it does not change the general shape of the distribution.

3.2.4. Glycemic-Range Distribution

As the central part of the EDA for this thesis, glycemic ranges distribution of REPLACE-BG dataset was also studied. As for a reference, the same ranges defined in Table 3.5 are used. Results are presented summarised in Table 3.13

Table 3.13: Glycemic range distribution in REPLACE-BG.

Glycemic range or category	Count	Percentage
Severe hypoglycemia	94,499	0.91%
Hypoglycemia	291,856	2.80%
Normoglycemia	6,636,660	63.62%
Hyperglycemia	2,437,218	23.36%
Severe hyperglycemia	972,175	9.32%
TBR (< 70 mg/dL)	386,355	3.70%
TIR (70–180 mg/dL)	6,636,660	63.62%
TAR (> 180 mg/dL)	3,409,393	32.68%

Figure 3.7 makes the imbalance towards clinically critical glycemic ranges clearly visible. Quantitatively, as described in Table 3.13, normo-glycemia is the dominant range representing 63.62% of all valid glucose measurements, hyperglycemia and severe hyperglycemia are also very present – with TAR accounting for 32.68% of the measurements, while hypoglycemic ranges are much smaller in the graph, accounting for 3.70% of the total. This graph then provides visual confirmation of the under-representation of hypoglycemic

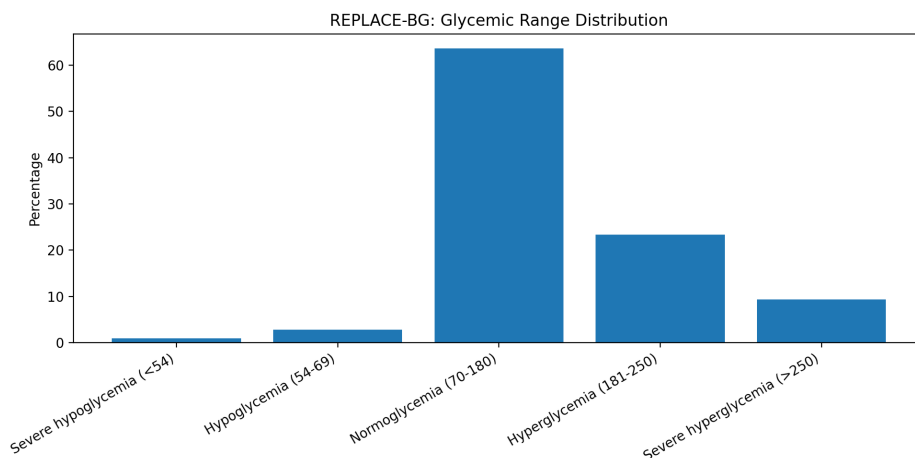


Figure 3.7: Glycemic range distribution

regions compared with normoglycemia and hyperglycemia.

Table 3.14: Imbalance ratios in REPLACE-BG.

Ratio	Value
Normoglycemia / hypoglycemia	17.18
Normoglycemia / severe hypoglycemia	70.23
Normoglycemia / severe hyperglycemia	6.83
Majority range / smallest minority range	70.23
TIR / TBR	17.18
TIR / TAR	1.95

Moreover, the concrete imbalance ratios were also quantified in Table 3.14. For example normoglycemic samples appear 17.18 times more frequently than hypoglycemic samples and 70.23 times more frequently than severe hypoglycemic samples.

3.2.5. Patient-Level Glycemic Distribution

The glycemic ranges distribution was also computed independently for each patient in order to make a more horizontally representative analysis. A summarized version of the results is presented in Table 3.15

Patient-level variability is also presented in the form of a box-plot in Figure 3.8.

TBR boxplot is also concentrated very close to zero, with a median of 3.24% and a maximum value of 14.20%. This indicates that hypoglycemic measurements are mostly consistently rare for most patients. On the other hand, TIR and TAR show much wider variability: TIR has a median of 63.18%, but ranges from 16.94% to 96.77%, while TAR has a median of

Table 3.15: Patient-level glycemic range statistics in REPLACE-BG.

Metric	Value
Mean patient TBR	3.70%
Median patient TBR	3.24%
Minimum patient TBR	0.01%
Maximum patient TBR	14.20%
Mean patient TIR	63.37%
Median patient TIR	63.18%
Minimum patient TIR	16.94%
Maximum patient TIR	96.77%
Mean patient TAR	32.94%
Median patient TAR	33.91%
Minimum patient TAR	0.51%
Maximum patient TAR	82.00%

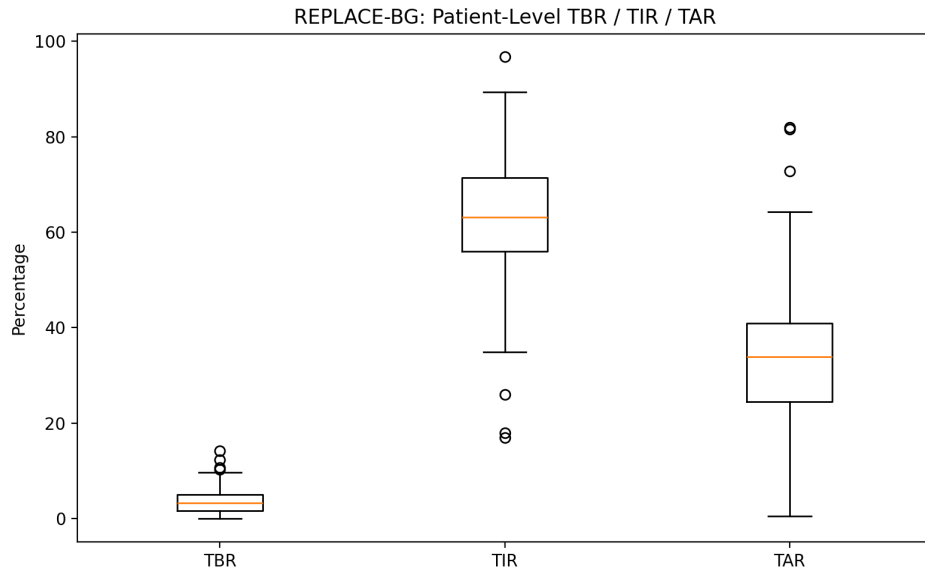


Figure 3.8: Glucose distribution in REPLACE-BG.

33.91% and ranges from 0.51% to 82%. In any case, the plot confirms the consistent under-representation of hypoglycemic values across patients in this dataset too.

In this case, it is important to highlight the substantial degree of variability between patients, especially for TIR and TAR. The wide spread of the boxplots and the presence of several outliers indicate that, although normoglycemia is still the dominant range for most individuals, the proportion of time spent in range or above range differs considerably across the population. In practical terms, this means that REPLACE-BG shows a consistent global imbalance against hypoglycemia, but patient glycemetic profiles present a high degree of heterogeneity with respect to TIR and TAR.

3.2.6. Summary of Findings

REPLACE-BG contains 223 patients and 10,432,408 valid glucose measurements. The dataset covers the period from 2000-01-13 00:01:53 to 2000-11-25 14:39:43, with a total recording span of 317 days.

It shows a very homogeneous patient contribution: the largest patient contributes only 0.53% of all measurements, the top 10 patients account only for 5.08% of the total, and 104 out 223 patients are required to accumulate 50% of the dataset. This indicates that the global glucose distribution is less likely to be dominated by a small subset of patients.

The glucose distribution analysis shows a normo-diabetes pattern.

The imbalance ratios quantitatively confirms the glycemetic ranges imbalance. They show that normoglycemic samples appear 17.18 times more frequently than hypoglycemic samples and 70.23 times more frequently than severe hypoglycemic samples.

At patient level, as expected, TBR remains consistently low across patients. However, it is important to highlight the substantial variability across patients population in TIR and TAR: the minimum and maximum in both ranges can differ up to 80% from one patient to another.

3.3. T1DiabetesGranada Dataset

Since the subsequent analyses for REPLACE-BG are analogous to the ones performed for both Diatrend and REPLACE-BG, some of the EDA methodology has been also omitted here.

Like Diatrend, REPLACE-BG dataset includes both a patient-level information table that provides demographic information, and a glucose measurement table with the timestamped glucose values. Tables' fields are also analogous to those described for Diatrend 3.1.1. The main descriptive characteristics of the dataset are summarized in Table 3.9

3.3.1. Dataset Description

In the processed version used in this thesis, only the glucose measurement table was available. Unlike DiaTrend 3.1.1 and REPLACE-BG 3.3, no separate patient metadata parquet file was available for this dataset. Therefore, demographic variables such as age and sex could not be described for this cohort. In any case, measurement table's fields are also analogous to those described for Diatrend and REPLACE-BG. The main descriptive characteristics of the dataset are summarized in Table 3.16.

Table 3.16: General description of the T1DiabetesGranada dataset.

Metric	Value
Number of rows	30,477,434
Number of columns	7
Number of patients	643
Total number of glucose measurements	22,233,264
Start date	2018-01-06 13:39:00
End date	2022-03-21 21:00:00
Total recording span	1,535 days
Number of days with measurements	1,536
Rows with 5-minute alignment	0
Rows with 15-minute alignment	30,113,515

Because no separate patient metadata file was available, no metadata consistency analysis was performed for this dataset. All subsequent analyses are based exclusively on the 643 patients identified from the measurement table.

This initial description provides the basic context needed to interpret the following analyses.

3.3.2. Patient Contribution Analysis

For each patient, the number of glucose measurements and the number of days with at least one measurement were computed, and summarised in Table 3.17

Figure 3.9 shows a clear imbalance in the number of measurements contributed by each patient in T1DiabetesGranada. Although no single patient dominates the dataset individually, the graph reveals a steep decline in the distribution followed by a long tail of patients with progressively fewer measurements. This indicates that measurements availability is highly heterogeneous across the 643 patients population.

Quantitatively, as shown in Table 3.17, the largest patient contributes only 0.62% of all measurements and the top 10 patients account for 5.48%, which may seem low when considered in isolation by simply looking at the graph, but is actually partly explained by the large number of patients in

Table 3.17: Patient contribution summary in T1DiabetesGranada.

Metric	Value
Mean measurements per patient	34,577.39
Median measurements per patient	28,223
Minimum measurements per patient	3,058
Maximum measurements per patient	137,293
Mean days per patient	391.24
Median days per patient	322
Minimum days per patient	33
Maximum days per patient	1,463
Percentage contributed by the largest patient	0.62%
Percentage contributed by the top 5 patients	2.91%
Percentage contributed by the top 10 patients	5.48%
Number of patients contributing 50% of the dataset	158

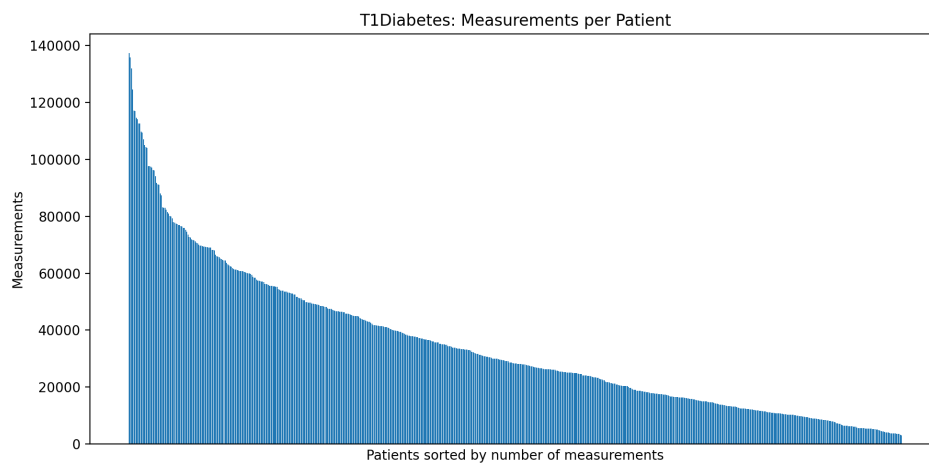


Figure 3.9: Measurements per patient in T1DiabetesGranada.

the dataset. The fact that 158 out of 643 patients are needed to accumulate 50% of the measurements also means that the remaining 485 patients contribute the other half, which implicitly implies those many individuals having much shorter recording histories. Therefore, while T1DiabetesGranada is not dominated by a few extreme patients – like DiaTrend, it is still very imbalanced in terms of "per-patient data availability".

3.3.3. Glucose Descriptive Statistics

The global distribution of glucose values was also analyzed and summarised in Table 3.18.

Table 3.18: Glucose descriptive statistics in T1DiabetesGranada.

Metric	Value
Minimum glucose	40 mg/dL
Maximum glucose	500 mg/dL
Mean glucose	164.48 mg/dL
Median glucose	153 mg/dL
Standard deviation	71.36 mg/dL

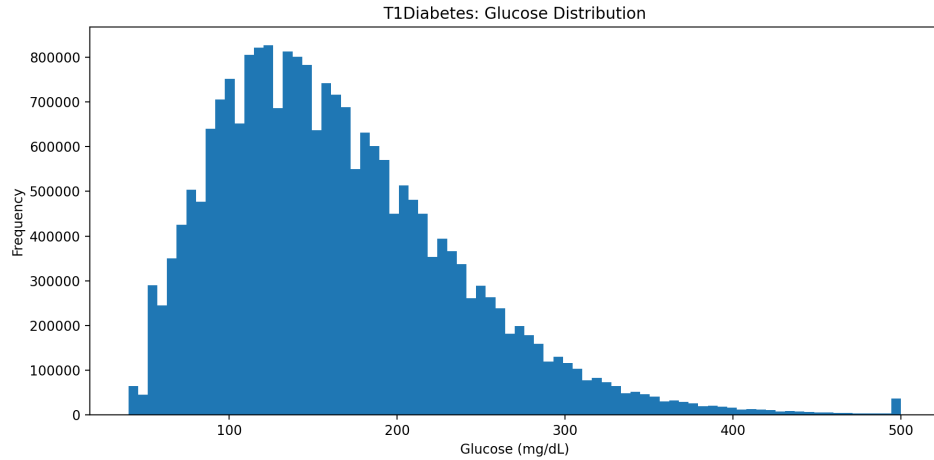


Figure 3.10: Glucose distribution in T1DiabetesGranada.

As shown in Figure 3.10, and also analogously to the findings for DiaTrend and REPLACE-BG, the glucose distribution follows a typical diabetes-related pattern.

Similarly to DiaTrend's and REPLACE-BG's, the histogram also presents an irregular, although less exaggerated, "one peak, one drop" pattern, that may be related to the measurement resolution. Once more, it does not change the general shape of the distribution.

3.3.4. Glycemic-Range Distribution

The most critical focus of this study is put in glycemic ranges distribution. As for a reference, the same ranges defined in Table 3.5 are used. The results of T1Diabetes' glycemic ranges distribution analysis are presented summarised in Table 3.19.

Table 3.19: Glycemic range distribution in T1DiabetesGranada.

Glycemic range or category	Count	Percentage
Severe hypoglycemia	220,610	0.99%
Hypoglycemia	840,722	3.78%
Normoglycemia	13,274,105	59.70%
Hyperglycemia	5,281,558	23.76%
Severe hyperglycemia	2,616,269	11.77%
TBR (< 70 mg/dL)	1,061,332	4.77%
TIR (70–180 mg/dL)	13,274,105	59.70%
TAR (> 180 mg/dL)	7,897,827	35.52%

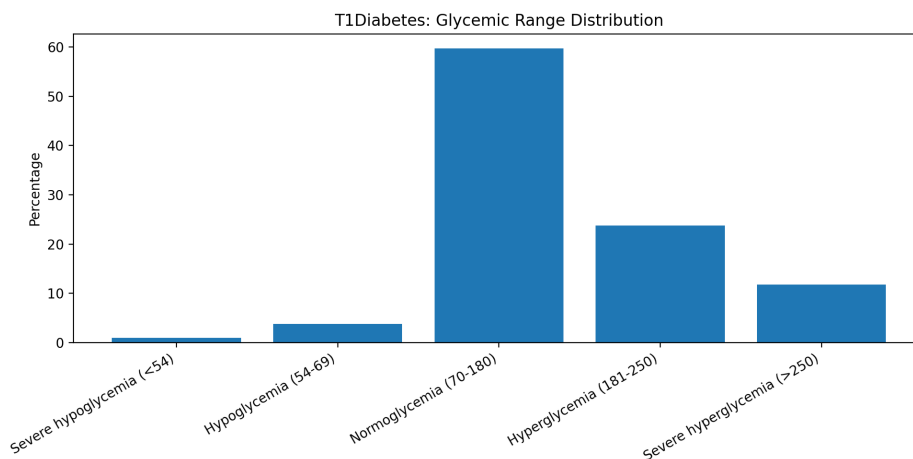


Figure 3.11: Glycemic range distribution

Figure 3.11 makes the imbalance towards clinically critical glycemic ranges clearly visible. Quantitatively, as described in Table 3.19, normo-glycemia is also the dominant range representing 59.70% of all valid glucose measurements, hyperglycemia and severe hyperglycemia are also very present – with TAR accounting for 35.52% of the measurements, while hypoglycemic ranges are much smaller in the graph, accounting for 4.77% of the total, and with severe hypoglycemia only accounting for 0.99% of the total valid values. This graph then provides visual confirmation of the expected under-representation of hypoglycemic regions compared with normoglycemia and hyperglycemia.

Table 3.20: Imbalance ratios in T1DiabetesGranada.

Ratio	Value
Normoglycemia / hypoglycemia	12.51
Normoglycemia / severe hypoglycemia	60.17
Normoglycemia / severe hyperglycemia	5.07
Majority range / smallest minority range	60.17
TIR / TBR	12.51
TIR / TAR	1.68

More specifically, imbalance ratios were also described in Table 3.20.

3.3.5. Patient-Level Glycemic Distribution

Per-patient glycemic ranges distribution has also been analyzed for T1DiabetesGranada, and results are presented in Table 3.21.

Table 3.21: Patient-level glycemic range statistics in T1DiabetesGranada.

Metric	Value
Mean patient TBR	4.45%
Median patient TBR	3.32%
Minimum patient TBR	0.00%
Maximum patient TBR	28.38%
Mean patient TIR	59.25%
Median patient TIR	60.05%
Minimum patient TIR	2.18%
Maximum patient TIR	97.72%
Mean patient TAR	36.30%
Median patient TAR	35.14%
Minimum patient TAR	0.00%
Maximum patient TAR	97.54%

As described in Table 3.21, once more, hypoglycemic measurements are consistently found to be rare for most patients, while TIR and TAR are the most common ranges but at the same time show much wider variability.

Patient-level variability is also presented as a box-plot in Figure 3.12. As in REPLACE-BG, the degree of variability between patients in this dataset, especially for TIR and TAR, is substantial: boxplots are wide-spread and several outliers are also present. This shows that, although normoglycemia is the dominant range for most individuals, the proportion of time spent in range or above varies up to 95% for some corner cases, reflecting considerable heterogeneity between patients.

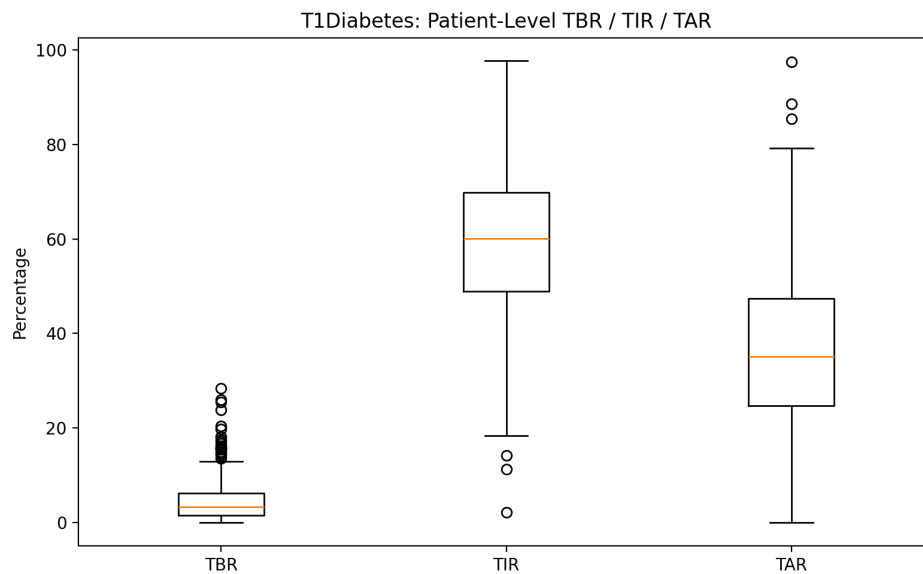


Figure 3.12: Glucose distribution in T1Diabetes Granada.

3.3.6. Summary of Findings

T1DiabetesGranada contains 643 patients and 22,233,264 valid glucose measurements. The dataset covers the period from 2018-01-06 13:39:00 to 2022-03-21 21:00:00, with a total recording span of 1,535 days.

The patient contribution analysis shows that T1DiabetesGranada is heterogeneous in terms of measurement availability, although it is not dominated by a very small group of patients: the largest patient contributes only 0.62% of all measurements, the top 10 patients account for 5.48%. and 158 out of 643 patients are required to accumulate 50% of the dataset, meaning that the remaining 485 patients contribute the other half with shorter monitoring histories.

The glucose distribution analysis describe the typically expected diabetes pattern.

Glycemic ranges imbalance is quantitatively confirmed in T1DiabetesGranada. For example, normoglycemic samples appear 12.51 times more frequently than hypoglycemic samples and 60.17 times more frequently than severe hypoglycemic samples.

At patient level, as expected, TBR remains consistently low across patients. However, as in REPLACE-BG, it is important to highlight the substantial variability across patients population in TIR and TAR: the minimum and maximum in both ranges can differ up to 95% from one patient to another as shown in Figure 3.12

3.4. Cross-Dataset Comparison

With all the three datasets analyzed separately, this section cross-compares them with each other, in order to determine whether glycemic ranges imbalance is consistent across the three of them, and to what extent.

Table 3.22: General comparison of the three datasets.

Metric	DiaTrend	REPLACE-BG	T1Diabetes Granada
Number of patients	51	223	643
Number of valid glucose measurements	7,667,356	10,432,408	22,233,264
Recording span	2,401 days	317 days	1,535 days
Mean measurements per patient	150,340.31	46,782.10	34,577.39
Median measurements per patient	80,488	48,688	28,223
Patients contributing 50% of the dataset	10	104	158

As shown in Table 3.22, the three datasets strongly differ on the number of patients, duration, and patient contribution. T1DiabetesGranada is the largest dataset in both number of patients and number of valid glucose measurements. REPLACE-BG has the most homogeneous per-patient contribution compared to the others. And DiaTrend, despite having the smallest number of patients, has the longest recording span, and as a result the highest mean number of measurements per patient.

Therefore, the most important difference appears in per-patient contribution. Both DiaTrend and T1Diabetes are relatively concentrated in a small subset of patients, since approximately only 19% and 26%, respectively, of the patients contribute 50% of the corresponding dataset. In contrast, REPLACE-BG requires approximately 46% of the patients to reach 50% of the measurements.

Table 3.23 confirms that the dominance of each of the 5 clinical glycemic ranges with respect to the others is repeated across the three datasets. Normoglycemia is always the majority range. TAR is also very present, although not as much as TIR. And TBR is consistently underrepresented in all datasets, with severe hypoglycemia also being the smallest glycemic range in all three datasets, remaining below 1% in all of them.

However, the strength of that dominance varies from one dataset to the other. REPLACE-BG has the highest TIR percentage, with 63.62% of its measurements in range, followed by T1DiabetesGranada with 59.70% and DiaTrend with 50.88%. TBR is lowest in DiaTrend, where it represents

Table 3.23: Cross-dataset comparison of glycemic ranges.

Metric	DiaTrend	REPLACE-BG	T1DiabetesGranada
Severe hypoglycemia	0.46%	0.91%	0.99%
Hypoglycemia	1.38%	2.80%	3.78%
Normoglycemia	50.88%	63.62%	59.70%
Hyperglycemia	28.60%	23.36%	23.76%
Severe hyperglycemia	18.68%	9.32%	11.77%
TBR	1.84%	3.70%	4.77%
TIR	50.88%	63.62%	59.70%
TAR	47.28%	32.68%	35.52%

only 1.84% of measurements, followed by REPLACE-BG with 3.70% and T1DiabetesGranada with 4.77%. The main difference between datasets appears in the hyperglycemic region. DiaTrend has the highest TAR percentage, with 47.28% of measurements above 180 mg/dL, including 18.68% in severe hyperglycemia. while REPLACE-BG and T1DiabetesGranada show lower TAR percentages.

An additional assessment of this imbalance is presented in Table 3.24 with the imbalance ratios of each dataset

Table 3.24: Cross-dataset comparison of imbalance ratios.

Ratio	DiaTrend	REPLACE-BG	T1DiabetesGranada
Normo / hypo	27.68	17.18	12.51
Normo / severe hypo	110.54	70.23	60.17
Normo / severe hyper	2.72	6.83	5.07
Majority / minority	110.54	70.23	60.17
TIR / TBR	27.68	17.18	12.51
TIR / TAR	1.08	1.95	1.68

These results show that the imbalance problem is not symmetric: low-glucose values, especially severe hypoglycemia, are consistently rare across all datasets, whereas high-glucose values are also less frequent than normoglycemia but present in higher proportions than low ones.

The patient-level comparison presented in Table 3.25 confirms that the imbalance patterns found globally are also largely present across individual patients and are not caused by asymmetric contributions from patients with different glycemic profiles. Across the three datasets, TBR generally remains low at the patient level, TIR is the majority range for most patients, and TAR represents a substantial but more variable proportion of measurements. Therefore, the dominance of normoglycemia and the underrepresentation of hypoglycemia are not driven by only a few large contributors but are broadly present at the patient level.

Table 3.25: Cross-dataset comparison of patient-level variability.

Metric	DiaTrend	REPLACE-BG	T1DiabetesGranada
Mean patient TBR	1.67%	3.70%	4.45%
Median patient TBR	1.18%	3.24%	3.32%
Mean patient TIR	53.97%	63.37%	59.25%
Median patient TIR	52.52%	63.18%	60.05%
Mean patient TAR	44.36%	32.94%	36.30%
Median patient TAR	46.50%	33.91%	35.14%
Patients with zero hypoglycemia	0	0	2
Patients with zero severe hypoglycemia	1	1	27
Patients with zero severe hyperglycemia	0	0	3
Patients where TIR is the majority range	48	219	608
Patients where TIR is the majority range (%)	94.12%	98.21%	94.56%

However, this is to be considered a general pattern, and does not necessarily imply that minority glycemic ranges are represented uniformly across every single individual. This is particularly evident in T1DiabetesGranada, where some patients contain no measurements in the hypoglycemic, severe hypoglycemic, or severe hyperglycemic ranges. Therefore, although the global and per-patient imbalance patterns mostly support the use of a common balancing strategy, the methodology should still remain sensitive to dataset-specific distributions and patient-level differences.

In summary, the comparison shows a consistent pattern across the three datasets: normoglycemia is the majority range, hypoglycemia is the clearest minority region, and severe hyperglycemia is underrepresented but less rare than hypoglycemia; which is also mostly present at patient-level.

3.5. Summary and Implications for Data Balancing

This chapter explored the structure and glucose distributions of DiaTrend, REPLACE-BG, and T1DiabetesGranada, in order to quantify the imbalance across clinically relevant glycemic ranges.

The results show that normoglycemia is the majority range in all three datasets. TIR represents 50.88% of DiaTrend, 63.62% of REPLACE-BG, and 59.70% of T1DiabetesGranada.

On the contrary, hypoglycemia is consistently the most underrepresented clinically relevant region. TBR represents only 1.84% of DiaTrend, 3.70% of REPLACE-BG, and 4.77% of T1DiabetesGranada. Severe hypoglycemia is even rarer, remaining below 1% in all datasets.

Severe hyperglycemia is also less frequent than normoglycemia in all

datasets, but it is not as rare as hypoglycemia. It represents 18.68% of DiaTrend, 9.32% of REPLACE-BG, and 11.77% of T1DiabetesGranada. This means that severe hyperglycemia should still be considered a minority region, but the imbalance against it is less extreme than the imbalance against hypoglycemia.

Therefore, although the exact proportion differs between datasets, the same general pattern is observed globally: most glucose measurements are concentrated in the normoglycemic range, low-glucose values are the clearest minority region, and hyperglycemia is also less represented than normoglycemia, but not as rare as hypoglycemia.

The patient-level analysis shows that, although the specific values for each patient have some natural degree of variability, the patterns found globally are also analogously present across population. TIR is the majority range for 94.12% of DiaTrend patients, 98.21% of REPLACE-BG patients, and 94.56% of T1DiabetesGranada patients. At the same time, the representation of minority ranges varies across individuals, especially in T1DiabetesGranada, where some patients have no measurements in specific minority glycemic ranges.

These findings have direct implications for data balancing. **First**, a balancing strategy is clearly justified to be approachable because all three datasets clearly show the expected glycemic-range imbalance. **Second**, hypoglycemia, severe hypoglycemia and severe hyperglycemia should receive particular attention in the balancing methodology, given their consistent rarity – although different from each other – and clinical importance. **Finally**, the methodology should not only be global, but also patient-aware: it should preserve individual patient distributions and glycemic profiles, and avoid generating samples for absent ranges.

In summary, the EDA confirms that the selected datasets are suitable for studying glycemic-range imbalance in Type 1 Diabetes CGM prediction. The observed imbalance is not merely theoretical, but empirically present across all three datasets, both globally and a patient level. The graphical and quantitative results provide the basis for defining the data balancing methodology in the next chapter, with the objective to increase the representation of clinically critical minority glucose regions without unnecessarily distorting the patient-level structure of the original data.

4. Experimental Methodology

This chapter defines the experimental methodology used to address the glycemic-range imbalance identified in Chapter 3

4.1. Experimental Methodology Proposal

4.1.1. Motivation from the Exploratory Data Analysis

The exploratory data analysis presented in Chapter 3 revealed a consistent imbalance across the three datasets under study. In all cases, the normoglycemic range represented the majority of the available measurements, while hypoglycemia, and particularly severe hypoglycemia, appeared only in a small fraction of the records. Severe hyperglycemia (>250 mg/dL) was also underrepresented compared to the normoglycemic range, although to a lesser extent. This imbalance was observed both at the global dataset level and at the patient level, suggesting that predictive models trained on these data may be biased towards the most frequently observed glycemic states.

4.1.2. Scope of the Balancing Process

First, it is important to note that, although the imbalance has been identified in the original CGM individual measurements, the balancing process will not be applied directly to these raw timestamped records. These measurements form actual patient-specific **time series** in which each glucose value is linked to previous and subsequent observations through a temporal dependency structure. Inserting, deleting, or duplicating individual records into this structure would create artificial trajectories that do not correspond to real physiological glycemic and introduce strong biases unrelated to the actual imbalance problem being studied. For this reason, balancing will be applied only after the original timestamped measurements have been transformed into the supervised **data windows** used by the predictive models. At this stage, each instance is represented as an input-output pair derived from the original CGM sequence, which makes it possible to apply resampling techniques without directly modifying the underlying timestamped measurements structure. This preserves the temporal consistency of the raw data while allowing the learning algorithm to be exposed to a less imbalanced distribution of glycemic ranges during model fitting.

Additionally, **balancing is applied only to the training data** because the purpose of these techniques is to modify the distribution of examples seen by the model during training, so that underrepresented glycemic ranges have a greater influence on parameter estimation. Therefore, validation and test sets must remain unchanged, since they are intended to preserve the original data distribution in order to provide an unbiased estimate of how the model behaves under realistic conditions. This decision is based on the standard methodological practice found in the imbalanced learning litera-

ture explored in Chapter 2. Throughout the state of the art reviewed in said chapter, resampling methods are consistently applied to the training data, while validation and test subsets are kept unmodified. Also, an additional specific review of methodological recommendations for this decision did not identify evidence suggesting that validation or test sets should be balanced. On the contrary, official documentation [25] explicitly warns that resampling the whole dataset before splitting, or equivalently resampling both training and test partitions, leads to data leakage and to evaluation on a distribution that no longer reflects the real use case. Similarly, the scikit-learn documentation [26] states that preprocessing operations must be learned only from the training data, and that test data should not be involved in fitting procedures.

4.1.3. Proposed Hybrid Balancing Strategy

To address the numerical imbalance limitation identified in Chapter 3, this study proposes a hybrid balancing strategy that combines synthetic oversampling of minority glycemic ranges, with controlled undersampling of the majority ranges. This approach preserves the total number of measurements in each dataset while redistributing them across glycemic ranges to achieve the desired proportions. This strategy will expose predictive models to a more representative set of clinically relevant glycemic events while preserving the overall characteristics of the original datasets.

4.1.4. Target Distributions

The exact proportions in which the hybrid approach described previously will change each dataset distribution are explored through three different target distributions: normalized baseline balancing, semi-balancing, and full balancing.

Normalized baseline

First, a normalized baseline target distribution is defined from the EDA results, providing a **single common reference** distribution across the three cohorts. This approach computes the target samples for each of the glycemic ranges as a sample-weighted mean of the three datasets percentages for that range, where each dataset contributes in proportion to its number of valid glucose samples. This decision avoids giving the same weight to datasets of very different size, as larger datasets should have a stronger influence on the common reference distribution.

This normalized baseline is obtained by pooling the glycemic-range counts of DiaTrend, REPLACE-BG, and T1DiabetesGranada, and then computing the global proportion of each range over the combined total.

Formally, let $n(d, r)$ denote the number of samples belonging to glycemic range r in dataset d , and let $N(d)$ denote the total number of valid glucose

samples in dataset d :

$$N(d) = \sum_r n(d, r).$$

The normalized baseline proportion for glycemic range r is then defined as:

$$p_{\text{norm}}(r) = \frac{\sum_{d \in \mathcal{D}} n(d, r)}{\sum_{d \in \mathcal{D}} N(d)},$$

where $\mathcal{D}$ contains DiaTrend, REPLACE-BG, and T1DiabetesGranada.

As a result, the normalized baseline target proportions are described in Table 4.1

Table 4.1: Normalized baseline target proportions.

Glycemic range	Percentage
Severe hypoglycemia	0.8688%
Hypoglycemia	3.0699%
Normoglycemia	59.0382%
Hyperglycemia	24.5741%
Severe hyperglycemia	12.4490%

To express the redistribution in absolute terms, Table 4.2 compares the original number of samples in each glycemic range with the number required to reach the normalized baseline distribution.

For dataset d and glycemic range r , the target number of samples is defined as:

$$n_{\text{target}}(d, r) = N(d) p_{\text{norm}}(r),$$

and the required adjustment is:

$$\Delta n(d, r) = n_{\text{target}}(d, r) - n(d, r).$$

A positive value indicates the number of samples that must be added through oversampling, whereas a negative value indicates the number of samples that must be removed through undersampling.

These values represent the theoretical redistribution of the complete datasets. There is some small residual differences caused by conversion to integer. These rounding differences will also be addressed in the implementation. By applying this re-distribution all the three datasets will share common proportions amongst the glycemic ranges, regardless of their total size.

Table 4.2: Absolute sample redistribution required to reach the normalized baseline distribution.

Dataset	Glycemic range	Original samples	Target samples	Adjustment	Operation
DiaTrend	Severe hypoglycemia	35,291	66,611	+31,320	Oversampling
DiaTrend	Hypoglycemia	105,621	235,383	+129,762	Oversampling
DiaTrend	Normoglycemia	3,901,129	4,526,670	+625,541	Oversampling
DiaTrend	Hyperglycemia	2,192,705	1,884,184	-308,521	Undersampling
DiaTrend	Severe hyperglycemia	1,432,610	954,508	-478,102	Undersampling
REPLACE-BG	Severe hypoglycemia	94,499	90,633	-3,866	Undersampling
REPLACE-BG	Hypoglycemia	291,856	320,268	+28,412	Oversampling
REPLACE-BG	Normoglycemia	6,636,660	6,159,107	-477,553	Undersampling
REPLACE-BG	Hyperglycemia	2,437,218	2,563,671	+126,453	Oversampling
REPLACE-BG	Severe hyperglycemia	972,175	1,298,729	+326,554	Oversampling
T1DiabetesGranada	Severe hypoglycemia	220,610	193,155	-27,455	Undersampling
T1DiabetesGranada	Hypoglycemia	840,722	682,547	-158,175	Undersampling
T1DiabetesGranada	Normoglycemia	13,274,105	13,126,120	-147,985	Undersampling
T1DiabetesGranada	Hyperglycemia	5,281,558	5,463,626	+182,068	Oversampling
T1DiabetesGranada	Severe hyperglycemia	2,616,269	2,767,816	+151,547	Oversampling

Fully Balanced Distribution

Full balancing consists of forcing all glycemic ranges to contain the same number of samples, thereby completely removing the imbalance between the different glycemic regions. Accordingly, for the five glycemic ranges, the target proportion is defined as:

$$p(r) = \frac{1}{5} = 20\%.$$

Thus, if (N) denotes the total number of training windows, each range is resampled to contain approximately $(N/5)$ samples. Ranges below this target are oversampled, whereas ranges above it are undersampled.

This approach is considered the main strategy against which the study premise will be evaluated.

Semi-Balanced Distribution

Semi-balancing is defined as an intermediate approach between the normalized baseline distribution and the fully balanced distribution. The aim of this strategy is to investigate the effects of reducing the imbalance without applying the more aggressive redistribution required by full balancing. However, this intermediate point is computed using a geometric mean rather than an arithmetic mean, corresponding to a midpoint on a logarithmic scale between the normalized baseline and the fully balanced distribution. The reason is that the objective is not to move each proportion halfway in absolute terms, but to reduce imbalance ratios halfway in relative terms.

Let $p_{\text{norm}}(r)$ denote the normalized baseline proportion of glycemic range r . Since the fully balanced proportion is:

$$p_{\text{full}}(r) = \frac{1}{5} = 0.20,$$

The unnormalized semi-balanced value for each range is then defined as:

$$q_{\text{semi}}(r) = \sqrt{p_{\text{norm}}(r) \cdot p_{\text{full}}(r)} = \sqrt{p_{\text{norm}}(r) \cdot 0.20}.$$

These values are then normalized so that their sum is equal to one:

$$p_{\text{semi}}(r) = \frac{q_{\text{semi}}(r)}{\sum_{k \in \mathcal{R}} q_{\text{semi}}(k)},$$

where $\mathcal{R}$ represents the set of five glycemic ranges.

As a result, applying these semi-balancing target proportions reduces the original imbalance ratios halfway on a logarithmic scale, while avoiding the stronger redistribution introduced by full balancing. The specific targets are shown in Table 4.3

Table 4.3: Semi-balanced target proportions.

Glycemic range	Target proportion
Severe hypoglycemia	4.9438%
Hypoglycemia	9.2934%
Normoglycemia	40.7547%
Hyperglycemia	26.2936%
Severe hyperglycemia	18.7145%

This approach has been considered because the literature also warns against applying balancing techniques too aggressively, especially when synthetic oversampling is used. Tarawneh et al. [27] argue that, in particularly sensitive domains like this one, oversampling methods may generate synthetic examples that do not truly represent the intended minority region, leading to poor real-world behaviour. Similarly, Alkhawaldeh et al. [28] also highlights that synthetic oversampling in medical datasets may introduce artificial instances that do not accurately represent the minority class, potentially producing unreliable results in real-world medical scenarios.

For these reasons, the three different strategies are experimentally explored: the normalized baseline provides a common sample-weighted EDA reference distribution, full balancing provides a naive upper reference where imbalance is completely removed, and semi-balancing provides an intermediate alternative.

4.1.5. Balancing Methods

The main balancing process will be performed using SMOTER [2] and SMOGN [3] because they are among the most established data-level methods specifically proposed for imbalanced regression problems, as stated in Chapter 2. While other imbalance-aware regression approaches exist, such

as SERA [19], Balanced MSE [20], or geometric variants [21], they either focus mainly on modifying the evaluation or loss function, or are less directly aligned with the objective of this thesis, which is to generate alternative balanced training sets while keeping the predictive models unchanged.

In addition to SMOTER [2] and SMOGN [3], a random resampling baseline will also be included in order to study a more naive approach. This method applies the same full-balancing and semi-balancing targets defined previously, but reaches them only through random duplication of minority-range windows and random removal of majority-range windows. Therefore it does not generate new synthetic glucose windows.

4.1.6. Patient-Aware Balancing

An additional methodological decision concerns whether balancing should be performed globally or independently for each patient. Since the EDA showed that patients do not share identical TBR, TIR, and TAR proportions, each individual presents a distinct glycemic profile shaped by their own physiological dynamics. Forcing every patient to reach the same balanced distribution would remove part of this heterogeneity and could generate patient profiles that do not reflect the original data.

For this reason, the proposed methodology will not apply strict per-patient balancing. Instead, the target distribution will be defined globally for each training set as defined in Chapter 4.1.4, while the resampling operations used to approach that target will be constrained by patient-level information. In other words, the objective is to balance the training set as a whole, not to force every patient to contain the same proportion of each glycemic range.

In practical terms, for each glycemic range, the number of samples to be added or removed will first be computed at global training-set level. When oversampling is required, only patients who already contain a minimum required amount of 2 samples in the corresponding range will be considered eligible. The required additional samples will then be distributed across these eligible patients, and in proportion to their original contribution to that range.

4.1.7. Fixed Experimental Configuration

In order to isolate the effect of the previously proposed balancing strategies, several experimental conditions will remain fixed throughout all experiments in order to provide a stable baseline for comparison. These decisions have been chosen because they represent common standard settings in glucose prediction studies and because the objective of this thesis is not to compare different predictive architectures, prediction horizons, or window configurations, but to evaluate how modifying the training-data distribution affects glucose prediction performance under equivalent modelling conditions.

For this reason, all experiments will use the same predictive algorithm,

namely a **Long Short-Term Memory** network. LSTM models are appropriate for this task because they are designed to process sequential data and can exploit temporal dependencies between previous CGM measurements.

The prediction horizon will be fixed at **one hour** into the future. Since the supervised windows are generated from 15-minute CGM samples, this corresponds to **predicting four samples ahead**. The input history length will be fixed at **eight previous measurements**, equivalent to **two hours of past glucose information**.

Model training is performed using five-fold cross-validation.

Finally, **Root Mean Squared Error** will be used as the main loss and evaluation metric.

4.1.8. Experimental Workflow

The experimental procedure is structured as a sequence of stages designed to isolate the effects of each methodological decision on glucose prediction performance. The workflow evaluates three balancing targets and the three balancing methods, while implementing patient-aware mechanisms and keeping the validation and test sets unchanged in all cases.

First, for each dataset, three training-set versions will be generated according to the normalized baseline target distribution, using each of the three proposed methods: random, SMOTER and SMOGN. The same fixed training configuration described in Section 4.1.7 will be used across all balanced training-set versions. Comparing the three methods under the same target distribution makes it possible to determine whether regression-aware synthetic oversampling provides benefits beyond basic random resampling.

Second, following the same procedure used for the normalized baseline, the three same balancing methods will be applied using the semi-balanced target distribution, in order to evaluate the effects of this intermediate balancing strategy.

Third, analogously to the previous steps, the same methods be applied using the fully balanced target distribution, in which all five glycemic ranges receive equal representation. This stage evaluates the effects of the most intensive balancing configuration and determines whether its potential improvements in clinically relevant minority ranges compensate for any degradation in overall predictive performance.

Finally, each resulting model will be evaluated using the same unmodified validation and test data. Performance will then be analysed using both numerical and clinical metrics, with particular to the minority regions identified during the EDA.

4.2. Statistical Analysis and Evaluation

The results of the experimentation will be statistically analysed in order to evaluate the study premise on a rigorous mathematical basis.

The general hypothesis states that **increasing the representation of rare glycemic ranges during training has a positive effect on predictive performance in clinically critical regions**. However, this formulation is too broad, since it englobes the full cohort of the problem. For the scope of this analysis, and in order to perform the evaluation through a single statistical test, the hypothesis is simplified to one primary clinical outcome. More specifically, the statistical analysis focuses on severe hypoglycemia, represented by TBR_2 , because it is the most clinically critical range as stated in the previous chapters.

4.2.1. Primary Clinical Outcome

Formally, the selected outcome for the statistical analysis is the global proportion of severe-hypoglycemia predictions located in Clarke Error Grid Zones A and B, denoted as:

$$AB_{TBR_2}. \quad (4.1)$$

Zone A contains predictions considered clinically accurate, whereas Zone B contains predictions that deviate from the reference glucose value but would not normally lead to an inappropriate clinical decision. Therefore, combined, they represent the percentage of predictions that remain **clinically acceptable**.

4.2.2. Primary Effect Measure

The reference baseline for all datasets is defined by the results of the corresponding **NORMALIZED BASELINE** experiment, since it uses a common targets distribution for the three of them. Then, the fully balanced configuration is compared with the corresponding normalized configuration.

For each balancing method m and cross-validation fold f , the effect is defined as:

$$\Delta AB_{TBR_2}^{(m,f)} = AB_{TBR_2}^{(m,f,\text{full})} - AB_{TBR_2}^{(m,f,\text{normalized})}, \quad (4.2)$$

where:

$$m \in \{\text{Random}, \text{SMOTER}, \text{SMOBN}\}. \quad (4.3)$$

and:

$$f \in \{1, 2, 3, 4, 5\}. \quad (4.4)$$

A positive value of $\Delta AB_{TBR_2}^{(m,f)}$ indicates that full balancing increases the proportion of clinically acceptable severe-hypoglycemia predictions relative to the normalized baseline for the corresponding method and fold. A negative value indicates the opposite.

The comparison is paired by balancing method and cross-validation fold: each fully balanced result is compared with its normalized counterpart obtained using the same method and fold.

The three method-specific differences are subsequently averaged within each fold to minimize the impact of the balancing method on the analysis:

$$\Delta AB_{\text{TBR}_2}^{(f)} = \frac{1}{3} \sum_{m=1}^3 \Delta AB_{\text{TBR}_2}^{(m,f)}. \quad (4.5)$$

The resulting five fold-level differences constitute the five observations used in the statistical test.

4.2.3. Statistical Hypothesis

Let μ_Δ denote the mean paired difference in AB_{TBR_2} between the fully balanced and normalized configurations across the five cross-validation folds, after averaging the corresponding differences across the three balancing methods.

Then, the null hypothesis states that full balancing does not produce a positive mean improvement relative to the normalized baseline:

$$H_0 : \mu_\Delta \leq 0. \quad (4.6)$$

The alternative hypothesis states that full balancing produces a positive mean improvement in severe-hypoglycemia predictions clinical acceptability:

$$H_1 : \mu_\Delta > 0. \quad (4.7)$$

Rejecting (H_0) with a (p)-value below the selected significance level would provide enough evidence for our research premise: that full balancing improves AB_{TBR_2} relative to the normalized baseline across all the evaluated balancing methods. If proven, this statistical evidence will then have to be evaluated together with any possible deterioration in other clinical metrics, and with the actual magnitude of μ_Δ . This is to determine whether the observed change is practically meaningful – *satisfying Equation 4.7 does not necessarily imply that the observed improvement is clinically relevant enough.*

4.2.4. Statistical Test

The primary hypothesis is evaluated using a one-sided paired Student's t -test. The paired design is appropriate because every fully balanced result is compared with the normalized result obtained using the same dataset and experimental configuration.

Let d_f denote the paired difference for each of the five folds as defined in Equation 4.5. The mean paired difference is then calculated as:

$$\bar{d} = \frac{1}{n} \sum_{f=1}^n d_f, \quad (4.8)$$

where $n = 5$ represents the five cross-validation folds.

The standard deviation of the paired differences is calculated as:

$$s_d = \sqrt{\frac{\sum_{f=1}^n (d_f - \bar{d})^2}{n - 1}}. \quad (4.9)$$

The paired Student's t -statistic is then defined as:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}. \quad (4.10)$$

Under the null hypothesis, this statistic follows a Student's t -distribution with:

$$n - 1 = 4 \quad (4.11)$$

degrees of freedom.

The significance level is established as:

$$\alpha = 0.01. \quad (4.12)$$

The null hypothesis will be rejected when:

$$p < \alpha. \quad (4.13)$$

The paired t -test assumes that the fold-level paired differences are approximately normally distributed. However, in this case, this assumption cannot be formally assessed, since only five fold-level differences are available for each dataset.

Additionally, since the same primary hypothesis is tested independently for DiaTrend, REPLACE-BG, and T1DiabetesGranada, the resulting p -values will also be studied using the Holm procedure [29] in order to also assess the probability of obtaining false-positive conclusions across the three dataset-specific tests.

4.2.5. Secondary Outcomes

The primary confirmatory analysis focuses exclusively on the difference in AB_{TBR_2} between full balancing and the normalized baseline. However, this outcome alone cannot describe the complete effect of modifying the training-data distribution.

The semi-balanced configuration is retained as an intermediate approach and will be presented descriptively.

Additionally, other regression and Clarke Error Grid metrics are also considered as secondary outcomes. These include:

- TBR RMSE and MAE;

- global RMSE, MAE, and A+B;
- TIR numerical and clinical performance;
- TAR numerical and clinical performance; and
- the individual proportions of predictions located in Clarke Error Grid Zones.

These secondary outcomes will be used to determine the impact of the primary clinical improvement – if proven – on other numerical prediction error metrics and on the movement of observations from clinically dangerous into acceptable ones. They are not treated as additional confirmatory hypotheses. They will mainly be interpreted through their numerical changes relative to the normalized baseline, and their consistency across balancing methods and datasets.

4.3. Balancing Implementation

The original training, validation, prediction, and evaluation pipelines had already been developed by the research team leading this thesis. Therefore, the objective is not to create a new prediction framework from scratch, but to incrust a new data-balancing mechanism into the existing framework so that the effect of different training-data distributions can be evaluated using the same training pipeline.

For this reason, the main contribution developed in this thesis is the BALANCING module. This module receives the training subset of each fold and generates a new version according to the selected balancing method and target distribution. Three methods were implemented: Random resampling, SMOTER and SMOGN. Each method can be combined with the normalized, semi-balanced, and fully balanced target distribution configurations described in Section 4.1.4.

The balancing process operates equivalently on the five clinically defined glycemic ranges. For each range, the number of required samples is calculated from the selected target distribution. Ranges containing more samples than required are undersampled, whereas minority ranges are oversampled: Random resampling generates additional observations by duplicating existing windows, while SMOTER and SMOGN generate synthetic windows from similar observations.

Additionally, for the synthetic approach (SMOTER and SMOGN), sample generation is patient-aware. The number of synthetic observations assigned to each patient is proportional to the number of eligible real observations available for that patient, and new windows are created using observations belonging to the same patient and glycemic range, rather than combining data from different patients.

Additionally, the inherited training pipeline was refactored to integrate this module. In each cross-validation fold, balancing is applied exclusively to the training subset before model fitting. The validation and test subsets remain unchanged so that model selection and final evaluation are performed using the original data distribution. The pipeline was also adapted to automatically execute the combinations of dataset, balancing method, and balancing level, while maintaining separate output directories and experiment records for every configuration.

All experiments use the same predictive model and training configuration. The model is a Long Short-Term Memory network with 128 recurrent units followed by a dense output layer that predicts a single future glucose value. It receives eight previous CGM measurements, representing two hours of glucose history, and predicts glucose four samples ahead, corresponding to a one-hour prediction horizon. The model is trained using the RMSE loss function and the Adam optimizer. Early stopping and model checkpointing are also used to retain the model with the best validation performance and avoid unnecessarily extending the training process.

A five-fold cross-validation procedure is also used for every experimental configuration. Results are calculated separately for the different glycemic ranges and are later weightedly-aggregated across folds. Consequently, the adapted pipeline provides a common and reproducible framework in which the only major experimental variation is the balancing strategy – method and target – applied to the training data.

Specific implementation can be checked in Appendix A.

4.4. Model Training

The complete experimental execution are run on the Calculus computing server. Training runs are submitted as batch jobs through **SLURM**.

This server-based workflow is necessary because of the large number of glucose windows and the computational cost of training all 27 – 3 datasets x 3 methods x 3 targets – balancing configurations across the different folds.

5. Experimental Results and Discussion

This chapter presents and discusses the results obtained from the experimental configurations described in Chapter 4. The analysis follows the evaluation framework established in Section 4.2, distinguishing between the primary confirmatory analysis and the secondary descriptive outcomes. The complete numerical results from all the 27 experimental configurations are provided in Appendix B

5.1. Balance Verification

Firstly, after generating the balanced training subsets, a brief verification was carried out to confirm that the balancing procedure lead to the intended redistribution of samples. This verification was conducted for every experimental configuration.

For each balanced subset, the achieved counts in severe hypoglycemia, hypoglycemia, normoglycemia, hyperglycemia, and severe hyperglycemia were compared with the theoretical target counts calculated by the balancing module. The same verification was performed for all 27 combinations of dataset, balancing method, and target distribution. The achieved counts matched the calculated targets "exactly". The only minor differences arose when proportional targets had to be converted into integer sample counts – the remaining samples caused by rounding were assigned to the normoglycemic range so that the final combined size remained correct. The complete logs where all the configurations targets appear can be found in Appendix B.

Once the correctness of the range-level counts had been confirmed, the resulting target-value distributions were inspected graphically.

***Note.** The REPLACE-BG dataset is used in this section as a representative example, since the corresponding distributions for the other datasets were also examined and confirmed to show analogous behaviour. The complete set of figures is provided in Appendix B. For each target configuration, the distributions obtained through random resampling, SMOTER, and SMOGN are presented together. This allows the effect of the balancing level to be distinguished from that of the sampling method. Although the three methods were configured to reach the same range-level counts, they differ in how samples are distributed within each glycemic range.*

Figure 5.1 presents the normalized configuration. This configuration introduces the smallest change, and the resulting distributions remain relatively close to the original glucose distribution, indicating that only limited oversampling or undersampling was required under this configuration. The three balancing methods produce very similar overall shapes, indicating that

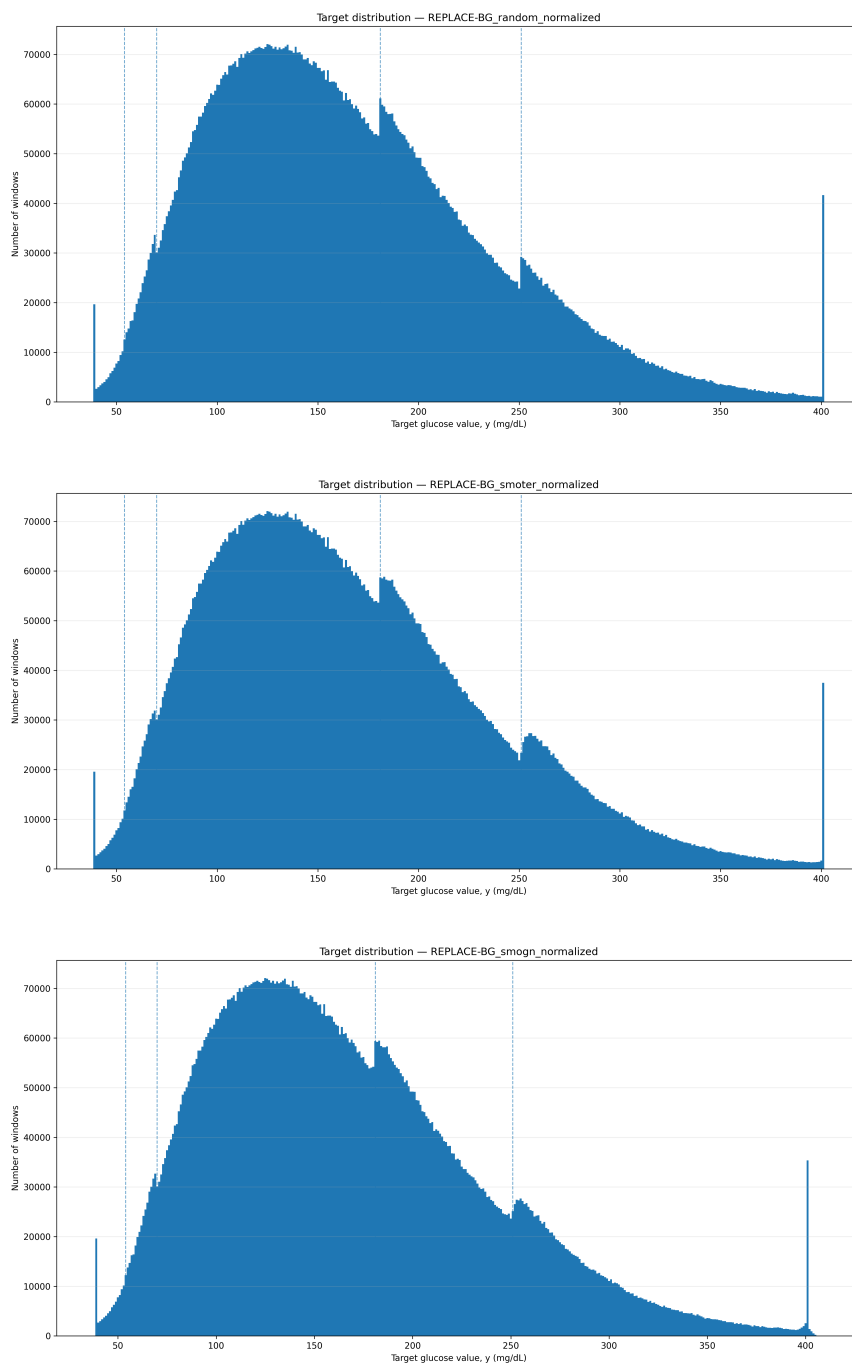


Figure 5.1: Balanced distribution on normalized target, methods comparison.

none of them introduces a systematic concentration of samples within particular regions of the glucose range.

Figure 5.2 presents the semi-balanced configuration. In this case, the redistribution is more noticeable: the minority ranges gain representation and the predominance of normoglycemia is reduced, while the overall distribution remains intentionally unequal. The configuration therefore represents an intermediate condition between the normalized and fully balanced distributions. The three methods also maintain comparable range-level shapes

Figure 5.3 presents the fully balanced configuration, which produces the strongest modification. The three methods also maintain comparable range-level shapes. Although each glycemic range contains the same total number of training windows, the five ranges have substantially different widths. Consequently, concentrating the same number of samples within a narrower interval produces much higher frequencies per glucose value, explaining the pronounced peaks in TBR.

Abrupt changes can also be observed at ranges limits: 54, 70, 181, and 251 mg/dL. These discontinuities are expected because each glycemic range is processed independently and receives a different sampling factor.

Overall, the verification confirmed that all balanced subsets matched their intended target counts. Also, it showed that the overall redistribution across the five glycemic ranges remains consistent, confirming that the balancing level determines the quantity of samples assigned to each range, while suggesting that the chosen balancing method hardly – does not almost – affects their internal distribution.

5.2. DiaTrend Results

5.2.1. Primary Statistical Test

As defined in Section 4.2, the statistical analysis evaluates whether full balancing improves the proportion of clinically acceptable predictions in severe hypoglycemia, AB_{TBR_2} , relative to the corresponding `NORMALIZED_BASELINE` configuration.

First, the difference between the fully balanced and normalized configurations was first calculated independently for each of the methods and cross-validation folds. Then, the three method-specific differences were averaged within each fold according to Equation 4.5. The resulting values, presented in Table 5.1 in percentage points, constitute the five fold-level observations used in the Student's t -test.

Full balancing produced a positive improvement in every cross-validation fold, with values that range from 47.54 to 79.62 percentage points. Moreover, all 15 individual method-fold comparisons were positive, which suggests that this effect is also consistent across balancing methods. The five observations used in the statistical test are therefore:

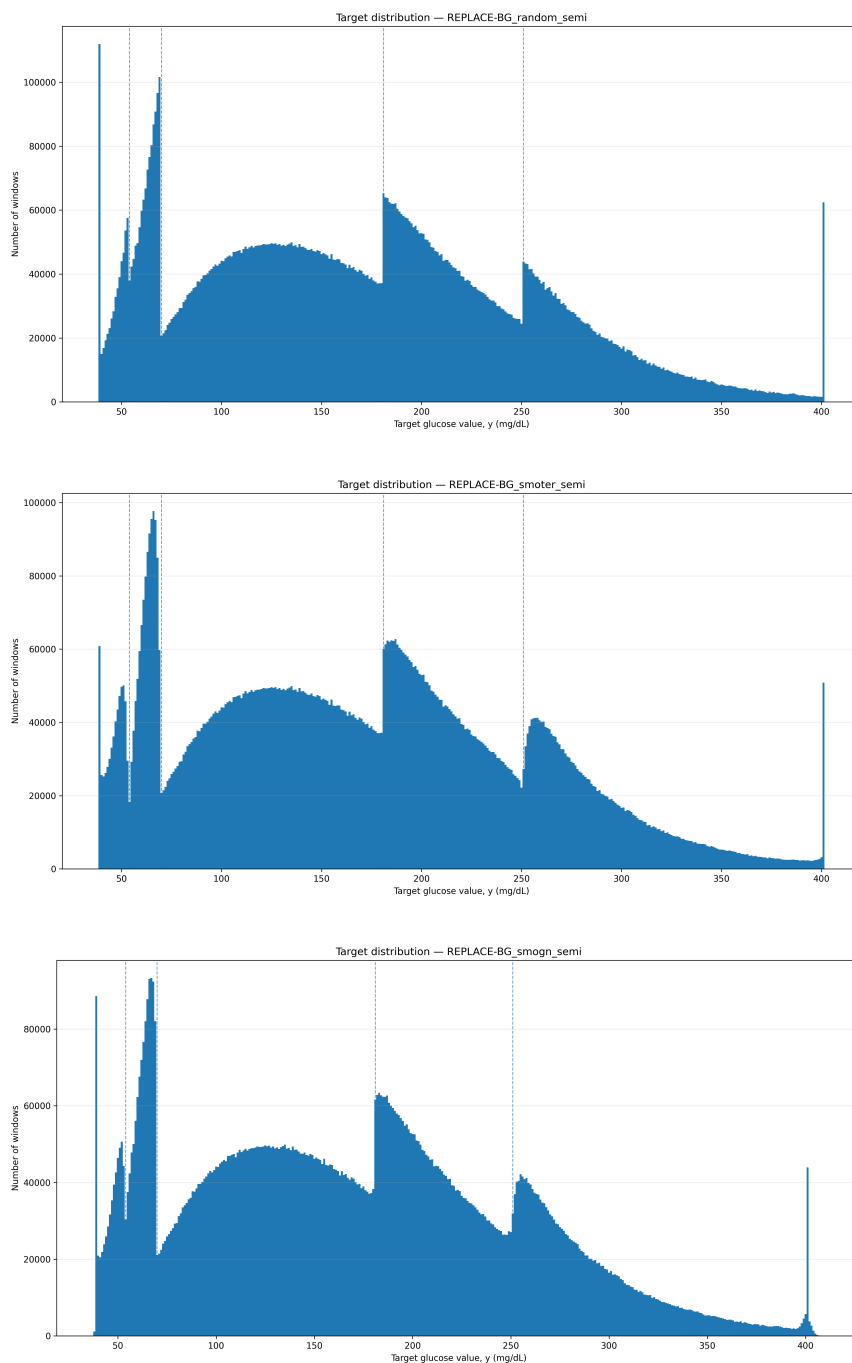


Figure 5.2: Balanced distribution on normalized target, methods comparison.

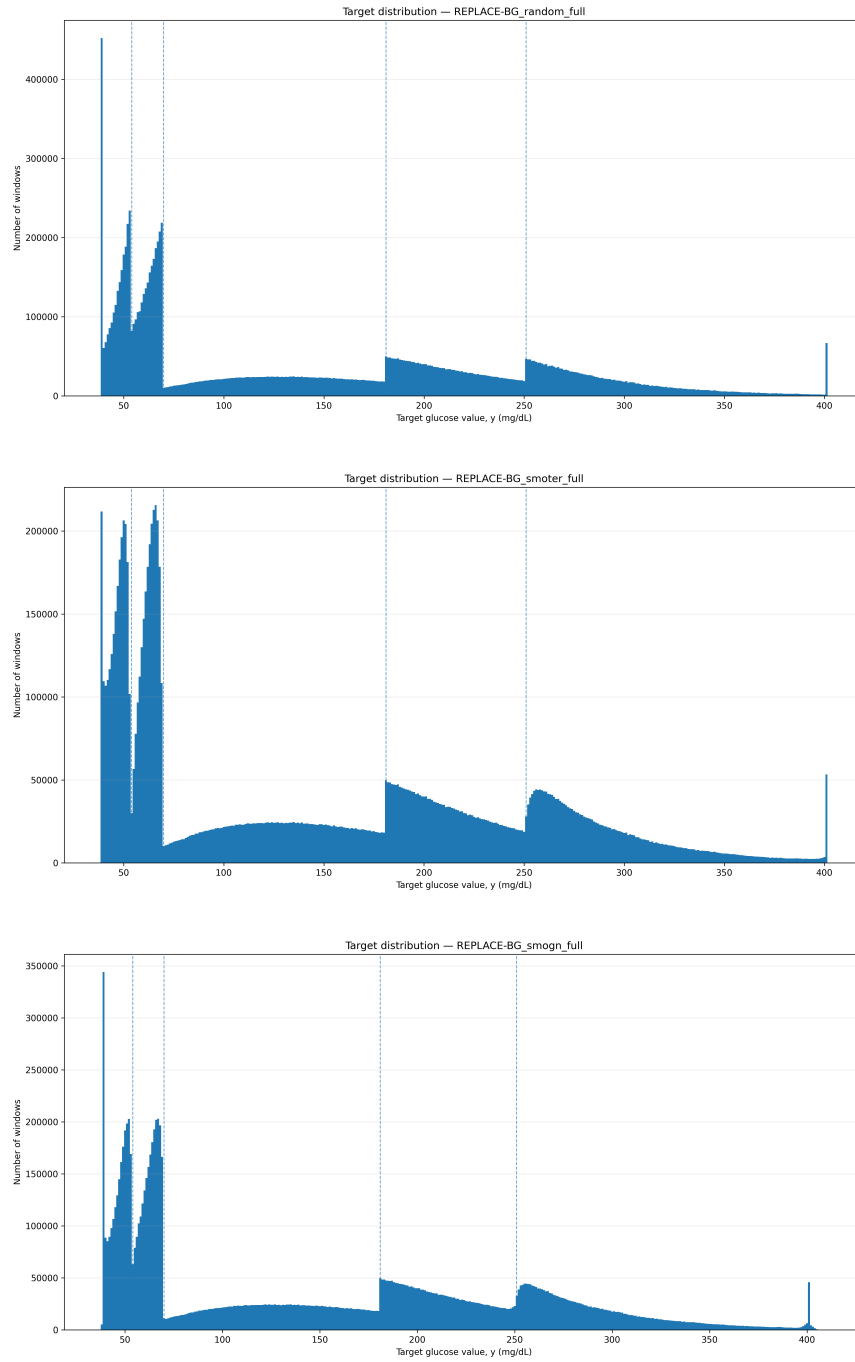


Figure 5.3: Balanced distribution on normalized target, methods comparison.

Table 5.1: DiaTrend fold-level results for the primary clinical outcome.

Fold	Normalized A+B	Full A+B	Δ A+B
1	0.36%	47.90%	+47.54 pp
2	1.17%	59.42%	+58.25 pp
3	6.71%	68.59%	+61.89 pp
4	0.02%	79.65%	+79.62 pp
5	7.55%	79.01%	+71.46 pp
Mean	3.16%	66.91%	+63.75 pp

$$d_f \in \{47.54, 58.25, 61.89, 79.62, 71.46\} \text{ pp.} \quad (5.1)$$

Their mean and standard deviation were:

$$\bar{d} = 63.75 \text{ pp,} \quad (5.2)$$

$$s_d = 12.33 \text{ pp.} \quad (5.3)$$

Applying the one-sided paired Student's t -test defined in Section 4.2.4 produced the following statistic:

$$t = \frac{63.75}{12.33/\sqrt{5}} = 11.57. \quad (5.4)$$

With $n - 1 = 4$ degrees of freedom, the corresponding one-sided p -value was:

$$p = 0.00016. \quad (5.5)$$

Since:

$$p < \alpha \quad (0.00016 < 0.01), \quad (5.6)$$

the null hypothesis defined in Equation 4.6 can be rejected for DiaTrend with a considerable degree of confidence.

Additionally, this statistical significance is accompanied by a substantial effect magnitude. Across the five folds, full balancing increased AB_{TBR_2} by an average of 63.75 percentage points: the mean value increased from 3.16% under the normalized baseline to 66.91% under full balancing.

The improvement was also consistent across the complete experimental configuration. Every fold showed a positive mean difference, and every individual comparison involving random resampling, SMOTER, or SMOGN favoured full balancing. Consequently, the result does not appear to depend on a particular balancing method or cross-validation partition.

Therefore, the primary statistical result supports the study hypothesis for DiaTrend: **increasing the representation of rare glycemic ranges through full balancing produces a statistically significant and clinically substantial improvement in the proportion of acceptable predictions in severe hypoglycemia**. The remaining secondary outcomes must subsequently determine whether this benefit is accompanied by acceptable changes in the other glycemic ranges.

5.2.2. Secondary Outcomes

Together with the primary statistical analysis, a secondary analyses was performed to study the impact of balancing in the rest of regression and clinical metrics. A single statistical test, narrowed to a very particular although critical metric, should not considered enough to determine the full effect.

***Note.** The secondary analysis uses the aggregate results obtained after combining the predictions from all folds for each experimental configuration. Consequently, the aggregate percentages reported in this subsection do not necessarily coincide exactly with the unweighted mean of the five fold-level percentages used in the previous statistical test. In any case, full results can be checked in Appendix B. Considering the large number of experiments and the corresponding performance metrics, for the scope of this secondary analysis, results are presented mostly summarized through their mean and descriptive standard deviation. Also, the normalized configuration remains the reference for interpreting the changes introduced by semi and full-balancing.*

Influence of the Balancing Method

Table 5.2 presents a condensed comparison between target distributions of some of the main global metrics. Each value represents the mean and standard deviation across **the three aggregate method results**.

Table 5.2: Main DiaTrend results.

Target	Entire RMSE	Entire A+B	TBR ₂ RMSE	TBR ₂ A+B	TBR ₁ RMSE	TBR ₁ A+B
Normalized	46.69 ± 0.10	95.23 ± 0.01	64.46 ± 0.97	3.74 ± 1.81	52.20 ± 0.41	1.60 ± 0.49
Semi-balanced	46.41 ± 0.14	96.02 ± 0.02	51.33 ± 0.58	35.73 ± 1.56	42.79 ± 0.33	28.99 ± 1.34
Fully balanced	49.92 ± 0.17	96.09 ± 0.03	41.07 ± 0.85	72.68 ± 0.35	35.14 ± 0.56	70.35 ± 0.25

Compared with the differences between target distributions, the variability across methods was consistently insignificant. For example, when analyzing global performance, the standard deviation across methods remained below 0.18 mg/dL for RMSE and below 0.03 percentage points for A+B. Under full balancing, TBR₂ A+B ranged only from 72.27% to 73.13%, and TBR₁ A+B ranged from 70.03% to 70.65%.

The only considerable "method-related" variability was found in the normalized hypoglycemic results, where the model is actually known to perform

poorly. But even in this case, the difference between methods remained considerably smaller than the improvements obtained by changing the balancing intensity.

These results do not necessarily prove that the three methods are equivalent, since no formal statistically test was performed. However, they clearly show that the selected target distribution – or balancing intensity – has a much higher influence in the predictive performance than the particular oversampling method.

Effect of Balancing Intensity in Hypoglycemia

The hypoglycemic results – which proved to have the most substantial improvements – are also analyzed across the normalized, **semi-balanced**, and fully balanced configurations in order to determine how the predictive performance changes as the degree of balancing increases.

Table 5.3: DiaTrend performance in the hypoglycemic ranges, averaged across balancing methods.

Target	Range	RMSE	MAE	A+B	Zone D	Zone E
Normalized	TBR ₂	64.46	59.07	3.74%	94.20%	2.05%
Semi-balanced	TBR ₂	51.33	39.29	35.73%	61.98%	2.28%
Fully balanced	TBR ₂	41.07	24.66	72.68%	25.16%	2.16%
Normalized	TBR ₁	52.20	46.48	1.60%	96.33%	2.06%
Semi-balanced	TBR ₁	42.79	30.94	28.99%	68.71%	2.29%
Fully balanced	TBR ₁	35.14	19.07	70.35%	27.39%	2.25%

Table 5.3 summarizes some critical metrics for TBR for each of the balancing targets. Normalized target is presented as a reference. Zones B and C contained no predictions in these ranges, so A+B is equivalent to Zone A.

Semi-balancing resulted in a substantial intermediate improvement, relative to the normalized baseline: TBR₂ RMSE decreased by 13.13 mg/dL and A+B increased by 31.99%; In TBR₁, RMSE decreased by 9.41 mg/dL and A+B increased by 27.39%. These clinical gains were mainly produced by a redistribution from Zone D into Zone A.

Full balancing intensified this effect: TBR₂ RMSE decreased by 23.40 mg/dL and A+B increased by 68.93%; In TBR₁, RMSE decreased by 17.07 mg/dL and A+B increased by 68.75%. The redistribution from Zone D into Zone A was also similarly pronounced.

Semi-balancing therefore provided a valid intermediate response, and full balancing produced a substantially larger improvement in both numerical accuracy and clinical acceptability. While not formally proven with the corresponding statistical test, these results suggest that the predictive performance improves progressively as the degree of balancing increases – at least

up to a fully symmetrical distribution.

$$AB_{\text{TBR}}^{(\text{normalized})} < AB_{\text{TBR}}^{(\text{semi})} < AB_{\text{TBR}}^{(\text{full})}. \quad (5.7)$$

Global and Time-in-Range Trade-Off

The improvement in hypoglycemia must also be interpreted together with its effect on global performance and TIR, which actually represents the most frequently observed glycemic range. Table 5.4 summarizes these results.

Table 5.4: Aggregate global and TIR performance in DiaTrend, averaged across balancing methods.

Target	Range	RMSE	MAE	A+B	Zone D	Zone E
Normalized	Entire	46.69	34.26	95.23%	4.45%	0.04%
Semi-balanced	Entire	46.41	33.55	96.02%	3.48%	0.13%
Fully balanced	Entire	49.92	36.23	96.09%	2.67%	0.69%
Normalized	TIR	29.13	21.36	99.50%	0.00%	0.00%
Semi-balanced	TIR	32.43	23.27	99.33%	0.00%	0.00%
Fully balanced	TIR	38.23	28.21	99.01%	0.00%	0.00%

Semi-balancing mostly preserved global numerical performance in comparison with the normalized baseline. Global RMSE decreased by 0.28 mg/dL and A+B increased by 0.79 percentage points. Therefore, its hypoglycemic improvements were not accompanied by a general deterioration in the aggregate metrics.

The main cost of semi-balancing appeared in TIR, where RMSE increased by 3.30 mg/dL. However, TIR A+B decreased by only 0.17 percentage points and still remained above 99%.

Full balancing introduced a more pronounced numerical trade-off. Global RMSE increased by 3.23 mg/dL and MAE by 1.97 mg/dL, relative to the normalized baseline. Nevertheless, global A+B increased by 0.86 percentage points.

The largest numerical deterioration of full balancing appeared in TIR, where RMSE was increased by 9.10 mg/dL and MAE by 6.85 mg/dL. Nevertheless, IR A+B decreased by only 0.49 percentage points and remained at 99.01%. Moreover, no TIR predictions entered Zones D or E.

The deterioration, both in TIR and globally, was therefore primarily numerical rather than clinical: while predictions in this ranges became less precise, they remained almost entirely within clinically acceptable regions.

Hyperglycemic Ranges Performance

The effects in TAR were also studied and found to be positive, although they slightly differ between moderate and severe hyperglycemia, as shown in Table 5.5.

Table 5.5: Aggregate DiaTrend performance in the hyperglycemic ranges, averaged across balancing methods.

Target	Range	RMSE	MAE	A+B	Zone D	Zone E
Normalized	TAR ₁	45.41	36.13	97.37%	2.54%	0.00%
Semi-balanced	TAR ₁	45.84	34.72	97.49%	2.13%	0.26%
Fully balanced	TAR ₁	50.64	36.82	95.84%	2.02%	2.02%
Normalized	TAR ₂	77.32	65.72	89.24%	10.76%	0.00%
Semi-balanced	TAR ₂	72.91	60.38	91.00%	8.96%	0.05%
Fully balanced	TAR ₂	73.21	59.22	90.89%	8.72%	0.39%

Semi-balancing had only a limited impact on TAR₁: RMSE increased by 0.43 mg/dL, and A+B increased by 0.11 percentage points. TAR₂ improved more clearly: RMSE decreased by 4.41 mg/dL, Zone D by 1.80 percentage points, and A+B increased by 1.76 percentage points.

Full balancing maintained most of the improvement in TAR₂. Relative to the normalized baseline, RMSE decreased by 4.11 mg/dL, and Zone D by 2.04 percentage points. A+B increased by 1.65 percentage points. In contrast, TAR₁ presented a localized deterioration: RMSE increased by 5.23 mg/dL and A+B decreased by 1.54 percentage points. Although Zone A increased by 2.05 percentage points and Zone D decreased by 0.52 percentage points, Zone E increased from zero to 2.02%. This adverse effect should therefore be considered in the overall interpretation of full balancing.

Overall, although some deterioration was introduced in TAR₁, it remained considerably smaller than the improvements achieved in TBR₂ and TBR₁. Moreover, severe hyperglycemia benefited from both semi-balancing and full balancing.

Clarke Error Grid Interpretation

Figures 5.4 and 5.5 present representative Clarke Error Grids for the semi-balanced and fully balanced random-resampling configurations. Having shown that the three balancing methods produce similar results, random resampling is selected for visualization because it represents the most simple approach.

These Clarke Error Grids are consistent with the quantitative analysis. They support the interpretation that the increase in global RMSE under balancing does not represent a uniform clinical deterioration. Instead, it mainly

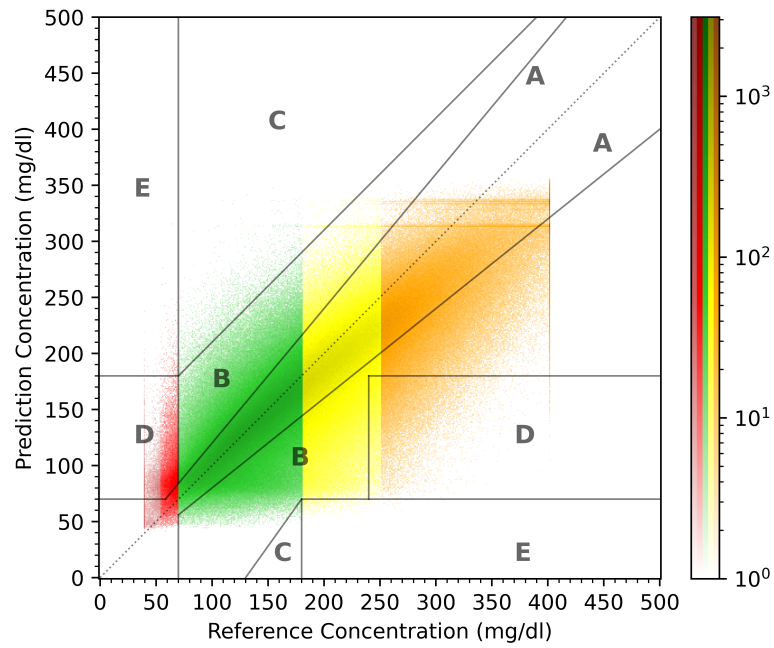


Figure 5.4: Clarke Error Grid for DiaTrend using random resampling and the semi-balanced target distribution.

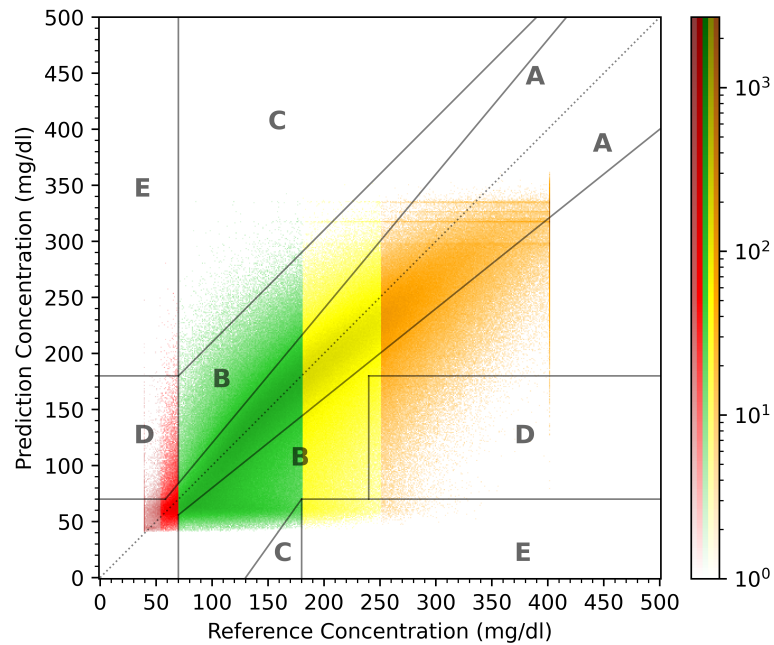


Figure 5.5: Clarke Error Grid for DiaTrend using random resampling and the fully balanced target distribution.

reflects reduced numerical precision in TIR, together with a substantial improvement in hypoglycemia.

5.2.3. DiaTrend Summary

The primary statistical analysis rejected the null hypothesis for DiaTrend. Using the fold-level analysis, full balancing increased AB_{TBR_2} by an average of 63.75 percentage points relative to the normalized baseline, with a one-sided p -value of 0.00016.

The aggregate secondary results support this conclusion. When predictions from all folds were combined, mean TBR_2 A+B across methods increased from 3.74% under normalized balancing to 72.68% under full balancing. TBR_1 A+B similarly increased from 1.60% to 70.35%. These improvements were accompanied by substantial reductions in RMSE, MAE, and Zone D.

The balancing target was found to have a much greater influence on predictive behaviour than the selected resampling method. Random resampling, SMOTER, and SMOGN produced closely aligned aggregate results under the same target distribution, although formal equivalence between them was not tested.

Semi-balancing provided a favourable intermediate response. It substantially improved both hypoglycemic ranges while preserving global RMSE and producing only a limited deterioration in TIR numerical accuracy.

Overall, full balancing is considered the most beneficial DiaTrend configuration. The moderate deterioration in global and TIR numerical accuracy is definitely outweighed by the magnitude and clinical relevance of the improvements obtained in TBR.

5.3. REPLACE-BG Results

Since the subsequent analyses for REPLACE-BG are analogous to the ones performed for DiaTrend, some parts of the analysis methodology have been omitted.

5.3.1. Primary Statistical Test

The same statistical analysis defined in Section 4.2 was also performed for REPLACE-BG. The resulting values are presented in Table 5.6 and constitute the five observations used in the paired Student's t -test.

Full balancing produced a positive improvement in every fold, with fold-level differences ranging from 73.54 to 78.96 percentage points. Additionally, all 15 individual method-fold comparisons were positive, with improvements ranging from 72.72 to 80.08 percentage points.

Therefore, the five observations used in the statistical test were:

$$d_f \in \{78.96, 78.62, 76.92, 76.00, 73.54\} \text{ pp.} \quad (5.8)$$

Table 5.6: REPLACE-BG fold-level results for the primary clinical outcome.

Fold	Normalized A+B	Full A+B	Δ A+B
1	0.01%	78.97%	+78.96 pp
2	0.00%	78.62%	+78.62 pp
3	0.06%	76.98%	+76.92 pp
4	0.00%	76.00%	+76.00 pp
5	0.00%	73.54%	+73.54 pp
Mean	0.01%	76.82%	+76.81 pp

Their mean and standard deviation were:

$$\bar{d} = 76.81 \text{ pp}, \quad (5.9)$$

$$s_d = 2.20 \text{ pp}. \quad (5.10)$$

Applying the one-sided paired Student's t -test produced the following statistic:

$$t = \frac{76.81}{2.20/\sqrt{5}} = 78.23. \quad (5.11)$$

With $n - 1 = 4$ degrees of freedom, the corresponding one-sided p -value was:

$$p = 8.00 \times 10^{-8}. \quad (5.12)$$

Since:

$$p < \alpha \quad (8.00 \times 10^{-8} < 0.01), \quad (5.13)$$

the null hypothesis defined in Equation 4.6 can be rejected for REPLACE-BG.

The statistical significance is also accompanied by a substantial and highly consistent effect magnitude. Mean AB_{TBR_2} increased from approximately 0.01% under normalized balancing to 76.82% under full balancing. Therefore, the REPLACE-BG results support the primary study hypothesis and show an even more consistent fold-level effect than that observed for DiaTrend.

5.3.2. Secondary Outcomes

The secondary outcomes analysis follows the same procedure used for DiaTrend.

Influence of the Balancing Method

Table 5.7 presents a condensed comparison of the principal global and hypoglycemic results.

Table 5.7: Main REPLACE-BG results.

Target	Entire RMSE	Entire A+B	TBR ₂ RMSE	TBR ₂ A+B	TBR ₁ RMSE	TBR ₁ A+B
Normalized	36.48 ± 0.12	95.08 ± 0.02	56.99 ± 0.47	0.01 ± 0.02	41.89 ± 0.43	0.33 ± 0.20
Semi-balanced	36.97 ± 0.05	96.42 ± 0.04	43.61 ± 0.29	27.08 ± 0.54	32.09 ± 0.09	36.90 ± 1.05
Fully balanced	41.18 ± 0.05	97.55 ± 0.04	32.57 ± 0.72	76.89 ± 0.85	24.92 ± 0.46	78.49 ± 0.65

As in DiaTrend, the differences between balancing methods were small compared with the changes introduced by the target distribution. The standard deviation of global RMSE remained at or below 0.12 mg/dL, while the standard deviation of global A+B remained below 0.05 percentage points, for both semi and full balancing targets distributions.

Therefore – although formal equivalence between methods is not demonstrated – the REPLACE-BG results also support the idea that balancing intensity has a much greater influence on predictive performance than any particular resampling method.

Effect of Balancing Intensity in Hypoglycemia

Table 5.8 presents the aggregate results for TBR₂ and TBR₁. As in DiaTrend.

Table 5.8: REPLACE-BG performance in the hypoglycemic ranges, averaged across balancing methods.

Target	Range	RMSE	MAE	A+B	Zone D	Zone E
Normalized	TBR ₂	56.99	53.96	0.01%	99.29%	0.69%
Semi-balanced	TBR ₂	43.61	36.13	27.08%	72.08%	0.84%
Fully balanced	TBR ₂	32.57	20.69	76.89%	22.22%	0.88%
Normalized	TBR ₁	41.89	38.18	0.33%	99.05%	0.62%
Semi-balanced	TBR ₁	32.09	23.27	36.90%	62.32%	0.78%
Fully balanced	TBR ₁	24.92	13.49	78.49%	20.69%	0.82%

Semi-balancing produced substantial improvements. Relative to the normalized baseline, TBR₂ RMSE decreased by 13.38 mg/dL and A+B increased by 27.06 percentage points. In TBR₁, RMSE decreased by 9.80 mg/dL and A+B increased by 36.56 percentage points.

Like in Diatrend analysis, full balancing intensified this effect. TBR₂ RMSE decreased by 24.42 mg/dL, while A+B increased by 76.88 percentage points. TBR₁ RMSE decreased by 16.97 mg/dL, and A+B increased by

78.16 percentage points. Once more, the clinical improvements were mainly produced by a movement from Zone D into Zone A.

The results therefore show the same descriptive relationship between balancing intensity in hypoglycemic performance, described by Equation 5.7

Global and Time-in-Range Trade-Off

Table 5.9 summarizes global and TIR performance after balancing the training sets.

Table 5.9: Aggregate global and TIR performance in REPLACE-BG, averaged across balancing methods.

Target	Range	RMSE	MAE	A+B	Zone D	Zone E
Normalized	Entire	36.48	26.71	95.08%	4.75%	0.02%
Semi-balanced	Entire	36.97	26.57	96.42%	3.29%	0.08%
Fully balanced	Entire	41.18	30.11	97.55%	1.66%	0.44%
Normalized	TIR	26.27	19.36	99.80%	0.00%	0.00%
Semi-balanced	TIR	29.57	21.30	99.71%	0.00%	0.00%
Fully balanced	TIR	35.89	27.05	99.46%	0.00%	0.00%

Semi-balancing increased global RMSE by only 0.49 mg/dL and reduced MAE by 0.15 mg/dL. At the same time, global A+B increased by 1.34 percentage points and Zone D decreased by 1.46 percentage points.

In TIR, semi-balancing increased RMSE by 3.30 mg/dL, but A+B decreased by only 0.09 percentage points and remained at 99.71%. The impact is also infimal.

Full balancing produced a larger numerical deterioration. Global RMSE increased by 4.70 mg/dL and MAE by 3.39 mg/dL. Nevertheless, global A+B increased by 2.47 percentage points, while Zone D decreased from 4.75% to 1.66%.

In this case, the largest numerical cost also appeared in TIR when full balancing, where RMSE increased by 9.62 mg/dL. However, TIR A+B decreased by only 0.34 percentage points and remained at 99.46%. No TIR predictions were located in Zones D or E.

Therefore, as in DiaTrend, the deterioration in TIR was mainly numerical rather than clinical. The global RMSE increase is strongly influenced by the reduced numerical precision in TIR, while global clinical acceptability actually improved slightly.

Hyperglycemic-Range Performance

Table 5.10 presents the results for moderate and severe hyperglycemia.

Table 5.10: Aggregate REPLACE-BG performance in the hyperglycemic ranges, averaged across balancing methods.

Target	Range	RMSE	MAE	A+B	Zone D	Zone E
Normalized	TAR ₁	42.35	33.35	98.29%	1.66%	0.00%
Semi-balanced	TAR ₁	42.26	31.74	98.37%	1.37%	0.20%
Fully balanced	TAR ₁	46.21	33.30	97.04%	1.29%	1.62%
Normalized	TAR ₂	65.80	55.27	91.62%	8.38%	0.00%
Semi-balanced	TAR ₂	61.65	50.37	93.13%	6.83%	0.03%
Fully balanced	TAR ₂	61.76	49.36	93.21%	6.50%	0.29%

Semi-balancing slightly improved TAR₁: RMSE decreased by 0.09 mg/dL, MAE by 1.61 mg/dL, and A+B increased by 0.08 percentage points. The improvement was clearer in TAR₂, where RMSE decreased by 4.15 mg/dL and A+B increased by 1.51 percentage points.

Full balancing maintained the TAR₂ improvement. Relative to normalized balancing, RMSE decreased by 4.04 mg/dL, MAE by 5.91 mg/dL, and A+B increased by 1.59 percentage points.

In contrast, TAR₁ showed a localized deterioration under full balancing. RMSE increased by 3.86 mg/dL and A+B decreased by 1.25 percentage points. Zone E also increased from zero to 1.62%.

This pattern is closely aligned with DiaTrend: severe hyperglycemia also benefited from balancing, whereas full balancing introduced a smaller adverse effect in moderate hyperglycemia.

Clarke Error Grid Interpretation

Figures 5.6 and 5.7 present representative Clarke Error Grids for the semi-balanced and fully balanced random-resampling configurations.

The grids support the range-specific results. Increasing the balancing intensity produces a substantial movement of hypoglycemic predictions from Zone D into Zone A. At the same time, the wider dispersion in the central glucose region is consistent with the increase in TIR RMSE.

As in DiaTrend, the increase in global numerical error therefore does not represent a uniform deterioration across the complete glucose spectrum. It is mainly associated with reduced numerical precision in TIR, together with a major improvement in the clinically critical hypoglycemic regions.

5.3.3. REPLACE-BG Summary

The primary statistical analysis rejected the null hypothesis for REPLACE-BG with a one-sided p -value of 8.00×10^{-8} . The improvement was also positive in every fold and in all 15 method-fold comparisons.

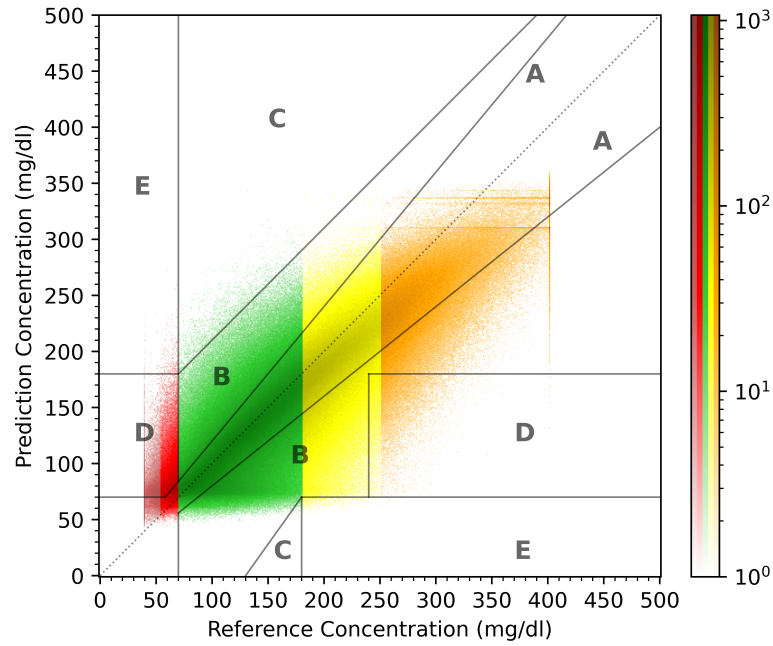


Figure 5.6: Clarke Error Grid for REPLACE-BG using random resampling and the semi-balanced target distribution.

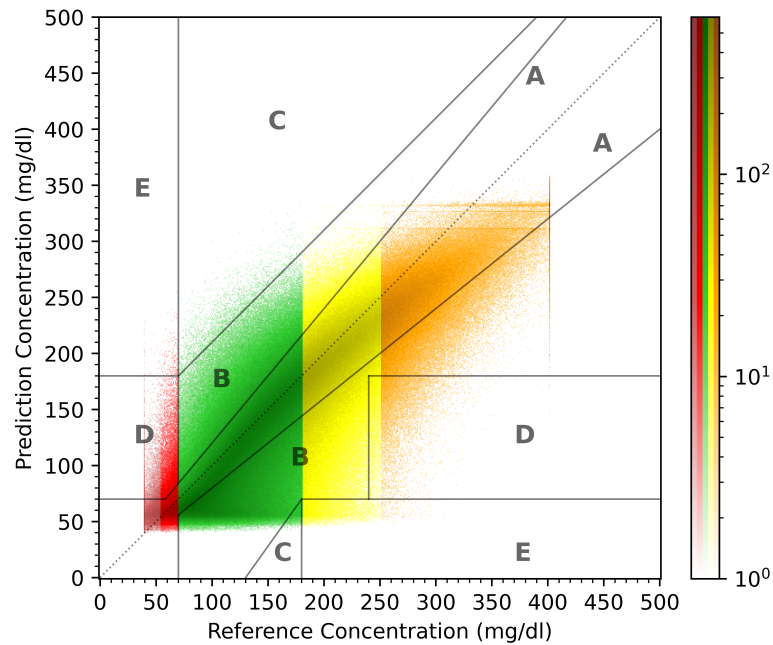


Figure 5.7: Clarke Error Grid for REPLACE-BG using random resampling and the fully balanced target distribution.

The aggregate results support this conclusion. Mean TBR_2 A+B increased from 0.01% under normalized balancing to 76.89% under full balancing, while TBR_1 A+B increased from 0.33% to 78.49%. These improvements were accompanied by substantial reductions in RMSE, MAE, and Zone D.

The balancing target again had a much greater influence than the selected resampling method. Semi-balancing provided an effective intermediate configuration.

Full balancing produced the strongest improvements in TBR_2 and TBR_1 . Its main cost was reduced numerical precision in TIR, although TIR A+B remained above 99% and no predictions entered Zones D or E. Severe hyperglycemia also improved, while moderate hyperglycemia presented a smaller localized deterioration.

Overall, full balancing is considered the most beneficial REPLACE-BG configuration. Its increase in global and TIR numerical error is considerably outweighed by the magnitude, consistency, and clinical relevance of the improvements in the hypoglycemic ranges.

5.4. T1DiabetesGranada Results

Part of the analysis methodology has been omitted since it is analogous to the one performed for both Diatrend and REPLACE-BG. Some results are presented and discussed directly.

5.4.1. Primary Statistical Test

Table 5.11 presents the resulting five fold-level observations arise from the primary statistical analysis.

Table 5.11: T1DiabetesGranada fold-level results for the primary clinical outcome.

Fold	Normalized A+B	Full A+B	Δ A+B
1	12.46%	88.85%	+76.39 pp
2	12.18%	89.32%	+77.14 pp
3	5.98%	90.16%	+84.18 pp
4	16.21%	88.61%	+72.40 pp
5	18.17%	87.69%	+69.53 pp
Mean	13.00%	88.93%	+75.93 pp

Full balancing produced a positive improvement in every fold, with fold-level differences ranging from 69.53 to 84.18 percentage points. All 15 individual method-fold comparisons were also positive, with improvements ranging from 67.60 to 90.53 percentage points.

The five observations used in the statistical test were:

$$d_f \in \{76.39, 77.14, 84.18, 72.40, 69.53\} \text{ pp.} \quad (5.14)$$

Their mean and standard deviation were:

$$\bar{d} = 75.93 \text{ pp,} \quad (5.15)$$

$$s_d = 5.55 \text{ pp.} \quad (5.16)$$

Applying the one-sided paired Student's t -test produced the following statistic:

$$t = \frac{75.93}{5.55/\sqrt{5}} = 30.60. \quad (5.17)$$

With $n - 1 = 4$ degrees of freedom, the corresponding one-sided p -value was:

$$p = 3.40 \times 10^{-6}. \quad (5.18)$$

Since:

$$p < \alpha \quad (3.40 \times 10^{-6} < 0.01), \quad (5.19)$$

the null hypothesis defined in Equation 4.6 can definitely be rejected for T1DiabetesGranada.

Full balancing increased mean AB_{TBR_2} from 13.00% to 88.93%, corresponding to an average improvement of 75.93 percentage points. The magnitude and consistency of this effect support the primary study hypothesis for T1DiabetesGranada.

5.4.2. Secondary Outcomes

Apart from the primary statistical test, an additional numerical analysis was performed on different secondary outcomes also for T1DiabetesGranada.

Influence of the Balancing Method

Table 5.12 summarizes the main global and hypoglycemic outcomes.

Table 5.12: Main T1DiabetesGranada results.

Target	Entire RMSE	Entire A+B	TBR ₂ RMSE	TBR ₂ A+B	TBR ₁ RMSE	TBR ₁ A+B
Normalized	37.80 ± 0.00	94.26 ± 0.04	40.83 ± 0.28	13.24 ± 1.07	36.82 ± 0.29	8.47 ± 1.07
Semi-balanced	37.68 ± 0.07	96.45 ± 0.03	28.32 ± 0.25	66.96 ± 0.79	29.40 ± 0.15	50.53 ± 0.50
Fully balanced	40.10 ± 0.21	97.53 ± 0.02	21.70 ± 0.26	88.87 ± 0.41	24.46 ± 0.17	79.06 ± 0.49

The variability across methods was again small relative to the differences between balancing targets. The standard deviation of global A+B remained

below 0.05 percentage points, while the standard deviation of global RMSE did not exceed 0.21 mg/dL.

Under full balancing, aggregate TBR₂ A+B ranged from 88.30% to 89.25%, while TBR₁ A+B ranged from 78.47% to 79.66%. Therefore – although formal equivalence was not tested – the results again suggest that predictive performance depends mainly on balancing intensity rather than on the particular resampling method.

Effect of Balancing Intensity in Hypoglycemia

Table 5.13 presents the aggregate results for TBR₂ and TBR₁.

Table 5.13: T1DiabetesGranada performance in the hypoglycemic ranges, averaged across balancing methods.

Target	Range	RMSE	MAE	A+B	Zone D	Zone E
Normalized	TBR ₂	40.83	36.87	13.24%	86.41%	0.35%
Semi-balanced	TBR ₂	28.32	19.11	66.96%	32.58%	0.47%
Fully balanced	TBR ₂	21.70	10.92	88.87%	10.62%	0.51%
Normalized	TBR ₁	36.82	31.86	8.47%	90.98%	0.54%
Semi-balanced	TBR ₁	29.40	19.78	50.53%	48.76%	0.71%
Fully balanced	TBR ₁	24.46	13.28	79.06%	20.14%	0.80%

Semi-balancing produced a particularly strong intermediate improvement. Relative to the normalized baseline, TBR₂ RMSE decreased by 12.52 mg/dL and A+B increased by 53.71 percentage points. In TBR₁, RMSE decreased by 7.42 mg/dL and A+B increased by 42.05 percentage points.

Full balancing produced the strongest results. TBR₂ RMSE decreased by 19.13 mg/dL, MAE by 25.95 mg/dL, and A+B increased by 75.63 percentage points. For TBR₁, RMSE decreased by 12.36 mg/dL, MAE by 18.58 mg/dL, and A+B increased by 70.59 percentage points.

These improvements were mainly explained by predictions moving from Zone D into Zone A.

As in the other datasets, the aggregate results show a progressive improvement as balancing intensity increases, suggesting the potential satisfaction of Equation 5.7

Global and Time-in-Range Trade-Off

Table 5.14 presents the global and TIR results.

Semi-balancing slightly improved global numerical performance. RMSE decreased by 0.12 mg/dL, MAE by 0.68 mg/dL, and global A+B increased by 2.19 percentage points.

Table 5.14: Aggregate global and TIR performance in T1DiabetesGranada, averaged across balancing methods.

Target	Range	RMSE	MAE	A+B	Zone D	Zone E
Normalized	Entire	37.80	27.55	94.26%	5.54%	0.03%
Semi-balanced	Entire	37.68	26.87	96.45%	3.17%	0.12%
Fully balanced	Entire	40.10	28.48	97.53%	1.75%	0.34%
Normalized	TIR	26.70	19.45	99.75%	0.00%	0.00%
Semi-balanced	TIR	29.65	21.17	99.61%	0.00%	0.00%
Fully balanced	TIR	34.13	24.82	99.39%	0.00%	0.00%

Full balancing increased global RMSE by 2.30 mg/dL and MAE by 0.93 mg/dL. However, global A+B increased by 3.26 percentage points, while Zone D decreased from 5.54% to 1.75%.

The largest numerical deterioration again occurred in TIR. Full balancing increased TIR RMSE by 7.43 mg/dL and MAE by 5.37 mg/dL. Nevertheless, TIR A+B decreased by only 0.36 percentage points and remained at 99.39%. Most of the redistribution occurred from Zone A into the still clinically acceptable Zone B, and no predictions entered Zones D or E.

Therefore, the T1DiabetesGranada results reproduce the same main trade-off: reduced numerical precision in TIR, but substantially improved clinical performance both globally and in hypoglycemia.

Hyperglycemic-Range Performance

Table 5.15 presents the results for TAR_1 and TAR_2 .

Table 5.15: Aggregate T1DiabetesGranada performance in the hyperglycemic ranges, averaged across balancing methods.

Target	Range	RMSE	MAE	A+B	Zone D	Zone E
Normalized	TAR_1	42.76	33.49	98.15%	1.77%	0.02%
Semi-balanced	TAR_1	41.94	31.32	98.12%	1.43%	0.37%
Fully balanced	TAR_1	43.59	31.06	97.46%	1.25%	1.22%
Normalized	TAR_2	66.22	55.81	92.59%	7.39%	0.00%
Semi-balanced	TAR_2	61.70	50.75	94.09%	5.85%	0.03%
Fully balanced	TAR_2	61.43	48.97	94.72%	5.14%	0.12%

Semi-balancing improved numerical performance in both hyperglycemic ranges. TAR_1 RMSE decreased by 0.82 mg/dL and MAE by 2.17 mg/dL, while A+B remained effectively stable. In TAR_2 , RMSE decreased by 4.52 mg/dL and A+B increased by 1.50 percentage points.

Full balancing produced a small deterioration in TAR_1 : RMSE increased by 0.83 mg/dL and A+B decreased by 0.69 percentage points. Zone E increased from 0.02% to 1.22%, although MAE still decreased by 2.43 mg/dL.

In contrast, TAR_2 improved further. RMSE decreased by 4.79 mg/dL, MAE by 6.84 mg/dL, and A+B increased by 2.13 percentage points. Therefore, balancing again benefited severe hyperglycemia while full balancing introduced a smaller localized deterioration in moderate hyperglycemia.

Clarke Error Grid Interpretation

Figures 5.8 and 5.9 present representative Clarke Error Grids for the semi-balanced and fully balanced random-resampling configurations.

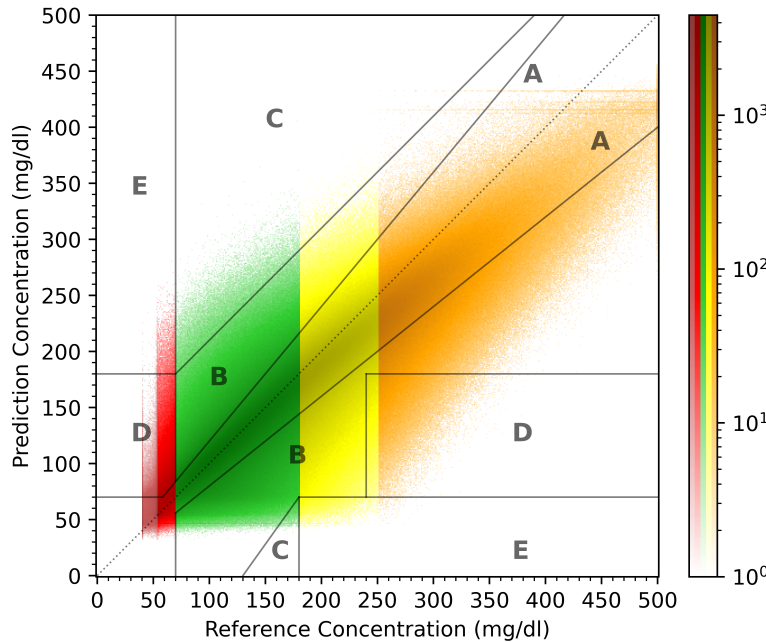


Figure 5.8: Clarke Error Grid for T1DiabetesGranada using random resampling and the semi-balanced target distribution.

The grids support the same interpretation observed in the previous datasets.

5.4.3. T1DiabetesGranada Summary

The primary statistical analysis rejected the null hypothesis for T1DiabetesGranada.

The aggregate results confirm the clinical magnitude of this effect. TBR_2 A+B increased from 13.24% under normalized balancing to 88.87% under full balancing, while TBR_1 A+B increased from 8.47% to 79.06%.

The balancing method again had little descriptive influence compared with the selected target distribution. Semi-balancing provided a strong in-

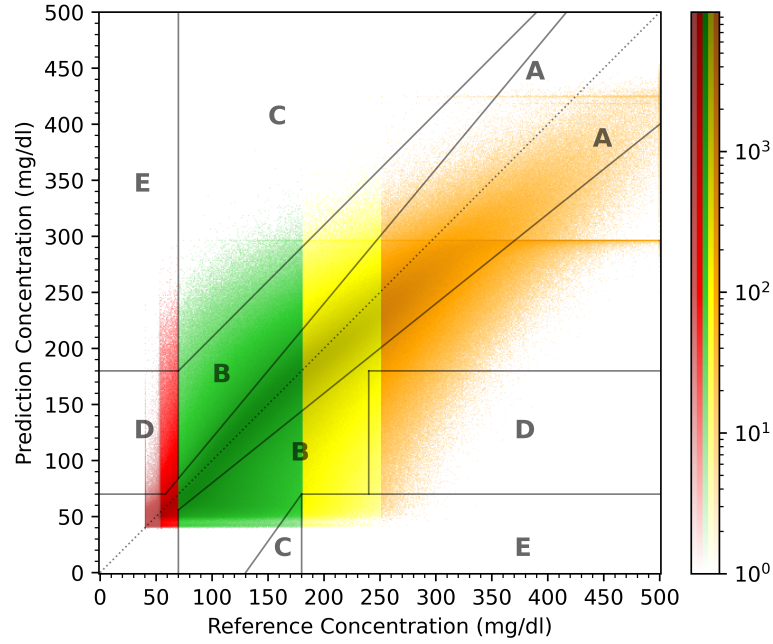


Figure 5.9: Clarke Error Grid for T1DiabetesGranada using random resampling and the fully balanced target distribution.

intermediate response and slightly improved global RMSE, while full balancing produced the largest improvements in both hypoglycemic ranges.

Full balancing reduced numerical precision in TIR, although TIR A+B remained above 99% and no predictions entered Zones D or E. Severe hyperglycemia also improved, while moderate hyperglycemia showed a smaller localized deterioration.

Overall, full balancing is considered the most beneficial T1DiabetesGranada configuration. Its moderate numerical cost in global performance and TIR is definitely outweighed by the magnitude and clinical relevance of the improvements obtained in TBR.

5.5. Cross-Dataset Statistical Synthesis

5.5.1. Comparison of the Primary Effects

Table 5.16 summarizes the primary statistical results across the three datasets.

Full balancing produced a statistically significant improvement in all three datasets. The effects obtained for REPLACE-BG and T1DiabetesGranada were nearly identical, whereas DiaTrend showed a slightly smaller and more variable, although still substantial, improvement.

Table 5.16: Cross-dataset comparison of the primary statistical outcome.

Dataset	$\overline{\Delta AB}_{\text{TBR}_2}$	t	p -value	Decision
DiaTrend	+63.75 pp	11.57	1.60×10^{-4}	Reject H_0
REPLACE-BG	+76.81 pp	78.23	8.00×10^{-8}	Reject H_0
T1DiabetesGranada	+75.93 pp	30.60	3.40×10^{-6}	Reject H_0

5.5.2. Holm-Adjusted Statistical Results

Since the primary hypothesis was evaluated independently in three datasets, the corresponding p -values were adjusted using the Holm procedure [29].

Table 5.17: Original and Holm-adjusted p -values for the primary outcome.

Dataset	Original p -value	Holm-adjusted p -value	Decision
DiaTrend	1.60×10^{-4}	1.60×10^{-4}	Reject H_0
REPLACE-BG	8.00×10^{-8}	2.40×10^{-7}	Reject H_0
T1DiabetesGranada	3.40×10^{-6}	6.80×10^{-6}	Reject H_0

All adjusted p -values still remained below the predefined significance level of $\alpha = 0.01$. Therefore, the correction did not alter any of the dataset-specific conclusions.

5.5.3. Consistency of the Primary Outcome

The primary effect was therefore consistent across the three datasets: full balancing improved AB_{TBR_2} in every fold and in all individual method–fold comparisons. Apart from the slightly smaller effect observed in DiaTrend, the results were analogous and consistently supported the primary hypothesis.

5.6. Cross-Dataset Secondary-Outcomes Synthesis

5.6.1. Influence of the Target Distribution

The three datasets showed the same general relationship between balancing intensity and predictive performance. Semi-balancing provided an intermediate response, and full balancing produced the largest clinical improvements. Therefore, the selected target distribution was consistently the principal factor determining the effect of balancing.

5.6.2. Influence of the Balancing Method

Random resampling, SMOTER, and SMOGN produced analogous results in all three datasets. The variability between methods remained considerably smaller than the differences between target distributions, indicating that balancing intensity had a substantially greater influence than the particular

resampling method.

5.6.3. Hypoglycemic Performance Across Datasets

Hypoglycemic performance improved progressively from normalized to semi-balanced and fully balanced configurations in every dataset, consistently reproducing the relationship suggested in Equation 5.7.

The main difference was observed in T1DiabetesGranada, where semi-balancing already produced a particularly strong improvement and full balancing reached the highest final TBR₂ A+B value. Nevertheless, full balancing produced substantial improvements in both TBR₂ and TBR₁ across all datasets, mainly through a redistribution of predictions from Zone D into Zone A.

5.6.4. Global Accuracy and Time-in-Range Trade-Off

The global and TIR trade-offs were also analogous. Semi-balancing largely preserved global numerical accuracy, whereas full balancing increased global and TIR errors. This deterioration was largest in REPLACE-BG and smallest in T1DiabetesGranada.

Despite the numerical deterioration, global A+B increased in every dataset. TIR A+B remained above (99%), and no TIR predictions entered Zones D or E. Consequently, the cost of full balancing was predominantly numerical, while its global clinical performance remained stable or improved.

5.6.5. Hyperglycemic Performance Across Datasets

The hyperglycemic results followed the same pattern across the three datasets. Semi-balancing had a neutral or slightly positive effect in TAR₁, while both semi and full balancing improved numerical and clinical performance in TAR₂.

Full balancing introduced a localized deterioration in TAR₁, including a small increase in Zone E. This effect was most pronounced in DiaTrend and smallest in T1DiabetesGranada. Nevertheless, it remained substantially smaller than the improvements obtained in the hypoglycemic ranges, while severe hyperglycemia consistently benefited from balancing.

5.7. Main Findings

The results provide a clear answer to the research question formulated in Chapter 2: *modifying the glycemic ranges distribution of the training data can substantially improve glucose prediction in underrepresented clinically critical regions*. This conclusion was reproduced in DiaTrend, REPLACE-BG, and T1DiabetesGranada, despite the particular differences between them, explored in Chapter 3.

- The exploratory data analysis demonstrated why this intervention was necessary. Severe hypoglycemia represented less than 1% of the avail-

able glucose measurements in all three datasets, while normoglycemia appeared between 60 and 110 times – depending on the dataset – more frequently. Also, it showed that this imbalance was not caused only by a small group of highly represented patients. It was also observed at the patient level, confirming that the limited representation of hypoglycemia is an intrinsic characteristic of real-world CGM data. This finding directly illustrated the limitation of evaluating glucose prediction models simply through a global regression loss. Since most training windows belong to TIR or moderate hyperglycemia, the model "plays it safe" by assuming all the measurements are in the normoglycemic range, in order to reduce its average error. From a purely mathematical perspective, this behaviour may appear reasonable. But from a clinical perspective, it means that a model can obtain satisfactory global performance while almost systematically failing in the regions where an accurate prediction is much more critical.

- By adjusting the glycemic ranges distribution to achieve a fully symmetrical balance, this behaviour changed enormously. The primary statistical outcome, $A + B_{\text{TBR}_2}$, increased by an average of 63.75 percentage points in DiaTrend, 76.81 percentage points in REPLACE-BG, and 75.93 percentage points in T1DiabetesGranada. The corresponding analogous one-sided paired Student's t -tests rejected the null hypothesis for the three datasets.
- The numerical magnitude of these effects is as important as their statistical significance. ¶The aggregate TBR_2 A+B percentage increased from 3.74% to 72.68% in DiaTrend, from 0.01% to 76.89% in REPLACE-BG, and from 13.24% to 88.87% in T1DiabetesGranada. Similar improvements were obtained for TBR_1 , where A+B reached 70.35%, 78.49%, and 79.06%, respectively. These gains were mainly due to a large movement of predictions from Zone D into Zone A.
- Full balancing produced an important transformation in models behaviour. The unbalanced original models were rarely capable of producing clinically acceptable predictions in hypoglycemia, whereas the fully balanced models correctly represented a large majority of them, and across three independent datasets.
- Additionally, the intermediate semi-balanced configuration also provided valuable evidence. In every dataset, its results were located between those of normalized and full balancing, which suggest a progression pattern – *not proven in this study*.

$$AB_{\text{TBR}}^{(\text{normalized})} < AB_{\text{TBR}}^{(\text{semi})} < AB_{\text{TBR}}^{(\text{full})}. \quad (5.20)$$

Nevertheless, semi-balancing approach was only investigated as an intermediate experimental condition rather than as the final preferred strategy.

- Another important finding is that the balancing target had a much greater influence than the oversampling method. Random resampling, SMOTER, and SMOGN produced very similar results for the same target distribution. In some cases, even simple random resampling produced larger improvements than other more complex synthetic methods. This result was not formally demonstrated.
- The improvements in hypoglycemia were accompanied by some numerical trade-offs. For example, full balancing increased global RMSE by 3.23 mg/dL in DiaTrend, 4.70 mg/dL in REPLACE-BG, and 2.30 mg/dL in T1DiabetesGranada. Similar deteriorations occurred in TIR. However, these changes must be interpreted according to where the additional error occurred. TIR A+B remained at 99.01% in DiaTrend, 99.46% in REPLACE-BG, and 99.39% in T1DiabetesGranada. No TIR predictions entered Clarke Error Grid Zones D or E. Therefore, these deteriorations only reflect a reduction in numerical precision but within clinically acceptable regions, rather than the introduction of clinically dangerous errors.
- At the global level, clinical acceptability actually improved in all three datasets despite the increase in RMSE. Global A+B increased from 95.23% to 96.09% in DiaTrend, from 95.08% to 97.55% in REPLACE-BG, and from 94.26% to 97.53% in T1DiabetesGranada. At the same time, the global proportion of Zone D predictions decreased considerably. These results demonstrate that a higher global RMSE does not necessarily indicate a clinically worse model.
- The most important finding of this study, from a clinical perspective, is that glucose prediction is not an abstract regression problem in which every error has the same consequence. Behind each prediction is a person who may need act before losing the ability to respond independently. The practical difference between detecting and failing to detect a severe hypoglycemic event may actually be the difference between an early manageable intervention and a situation that leads to physical impairment, seizure, loss of consciousness, or even death. For this reason, even a small improvement in this context would be worthy of serious consideration. However, the improvements observed in the present experiments were far from modest: the proportion of clinically acceptable severe-hypoglycemia predictions increased by approximately 64 to 77 percentage points in the primary fold-level analyses, and reached between 72.68% and 88.87% in the aggregate fully balanced results.

These results should not be interpreted as direct evidence that the evaluated models would actually perform satisfactorily in the real world. Such a conclusion would require ..., which is definitely not the aim of this study. However, the experiments clearly demonstrate that data balancing approaches can substantially reduce the type of predictive failures that would prevent a model from recognizing critical hypoglycemia events.

The broader conclusion is that glucose prediction models should not be optimized solely according to aggregate RMSE. When clinical impact is treated as a main priority, models can be substantially more useful and safer, even if they are slightly less accurate on average. Consequently, future glucose prediction systems should evaluate both the magnitude and the clinical location of their errors, and should explicitly consider the training-data glycemic ranges distribution as a fundamental decision.

5.8. Future Work

The results obtained in this thesis also open some possible directions for future research.

First, the target distribution could be extended beyond symmetrical full balancing. Equal representation of the five glycemic ranges was used in this study as a clear and naive approach, but it is not necessarily the optimal distribution. Future experiments could explore asymmetric targets that assign even greater presence to severe hypoglycemia. This would make it possible to identify the most clinically efficient distribution.

Second, the methodology should be evaluated with alternative predictive architectures: history lengths, prediction horizons, learning algorithms, etc.

Finally, data balancing could be combined with clinically aware loss functions that assign greater penalties to dangerous prediction errors. This would allow evaluating whether data-balancing and optimized loss functions are complementary to produce improvements in critical glycemic ranges.

6. Project Planning and Budget

This chapter presents the temporal planning and estimated economic cost for the execution of the project. The work was originally planned over thirteen weeks, from Week 0 to Week 12, with a total student workload of approximately 300 hours. In addition, one one-hour supervision meeting was scheduled once every two weeks.

6.1. Project Planning

The project was divided into six consecutive phases:

- **Weeks 0–3** were dedicated to the theoretical foundations: review of the state of the art, comprehension of the clinical context of Type 1 Diabetes, and study of clinical evaluation methods such as the Clarke Error Grid.
- **Weeks 4–5** focused on studying the structure the datasets in order to quantify their imbalance across glycemetic ranges.
- **Weeks 6–8** were dedicated to the definition and implementation of the balancing methodology.
- **Weeks 9–11** were planned for models training and results discussion
- **Week 12** was reserved for completing and reviewing the memory of the thesis and preparing the final submission.

6.1.1. Initial Workload Distribution

The planned hours were distributed according to the expected duration and complexity of each phase. The seven supervision meetings were included within the student’s total workload.

Table 6.1: Initial workload distribution.

Phase	Weeks	Planned hours
Theoretical foundations research	0–3	100
Dataset preparation and imbalance analysis	4–5	50
Data-balancing implementation	6–8	75
Model training and results discussion	9–11	75
Project closure	12	25
Supervision meetings	0–12	7
Total	0–12	332

This distribution corresponds to an average workload of 5 hours a day, 5 days a week, which results in approximately 25 hours per week.

6.1.2. Initial Temporal Planning

Table 6.2 summarizes the initial temporal distribution of the project using relative week numbers. The activities were organized sequentially, with model training and result analysis grouped within the same final experimental phase. Seven supervision meetings were planned at two-week intervals throughout the project.

Table 6.2: Initial temporal planning of the project.

Activity	W0	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Theoretical foundations research	•	•	•	•									
Dataset preparation and imbalance analysis					•	•							
Data-balancing implementation							•	•	•				
Model training and results discussion										•	•	•	
Project closure													•
Supervision meetings	•		•		•		•		•		•		•

6.2. Project Budget

This budget section estimates the economic cost of the personnel and computational resources required to complete the project.

6.2.1. Personnel Costs

The student workload was valued at 15 € per hour, using a rounded reference value based on the approximately 14 € per hour sakart reported by Glassdoor for Junior Data Scientist positions in Spain [30]. Applied to the planned 332 hours, this results in an estimated personnel cost of 4,980 €.

Academic supervision was estimated separately. One hour of supervision was planned for each of the seven project meetings. Using a rate of 30 € per hour, for each of the three tutors attending the meeting, the corresponding supervision cost was 630 €. The student’s participation in these meetings is already included within the 332-hour workload.

Table 6.3: Estimated personnel costs.

Personnel resource	Hours	Hourly rate	Estimated cost
Student work	332	15.00 €	4,980.00 €
Academic supervision	7 (x3)	30.00 €	630.00 €
Total personnel cost			5,610.00 €

6.2.2. Computational and Additional Costs

No server was purchased specifically for the project. The experimental execution used computing infrastructure provided by the university. How-

ever, its economic cost was simulated from an equivalent commercial cloud instance with one GPU and 32 GB of RAM. Specifically, an Amazon EC2 `g4dn.2xlarge` instance, equipped with one NVIDIA T4 GPU and 32 GiB of RAM, has an on-demand price of approximately \$0.752 per hour [31]. Therefore, a 24-hour experimental execution to train all the models was valued at approximately \$18.05, which is 15.65€.

All programming languages, libraries, machine-learning frameworks, and document-preparation tools used in the project were freely available. No paid subscriptions, electricity, Internet, printing, or additional material costs were included.

Table 6.4: Estimated computational and additional costs.

Cost category	Estimated cost
Cloud-equivalent GPU infrastructure	15.65 €
Software licences and subscriptions	0.00 €
Electricity and Internet	0.00 €
Travel and printing	0.00 €
Additional materials	0.00 €
Total computational and additional costs	15.65 €

6.2.3. Total Estimated Cost

The total estimated cost was obtained by adding the personnel and computational resource costs.

Table 6.5: Total estimated project budget.

Cost category	Estimated cost
Student personnel	4,980.00 €
Academic supervision	630.00 €
Computational infrastructure	15.65 €
Software and additional costs	0.00 €
Total estimated project cost	5,625.65 €

Therefore, the estimated economic value of the project was 5,625.65 €.

A. Source Code Repository

The source code developed for this project is available in the following GitHub repository:

`https://github.com/qillo/patient-heterogeneity-balancing`

The repository also contains the configuration files required to reproduce the experiments described in this thesis.

B. Experimental Results Repository

The complete experimental outputs generated during this project require approximately 8 GB of storage and are therefore hosted separately from the source-code repository.

The files are available at:

https://drive.google.com/drive/folders/1_8X2dCzbucMPxhzlpLq1J-ztBNfmjLwQ?usp=sharing

Please feel free to contact me at dsanortiz@correo.ugr.es for any further information.

Bibliography

- [1] Tadej Battelino et al. “Clinical Targets for Continuous Glucose Monitoring Data Interpretation: Recommendations From the International Consensus on Time in Range”. In: *Diabetes Care* 42.8 (June 2019), pp. 1593–1603. ISSN: 0149-5992. DOI: 10.2337/dci19-0028. eprint: <https://diabetesjournals.org/care/article-pdf/42/8/1593/529329/dci190028.pdf>. URL: <https://doi.org/10.2337/dci19-0028>.
- [2] Luís Torgo et al. “SMOTE for Regression”. In: *Progress in Artificial Intelligence*. Ed. by Luís Correia, Luís Paulo Reis, and José Cascalho. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, pp. 378–389. ISBN: 978-3-642-40669-0.
- [3] Paula Branco, Luís Torgo, and Rita P. Ribeiro. “SMOGLN: a Pre-processing Approach for Imbalanced Regression”. In: *Proceedings of the First International Workshop on Learning with Imbalanced Domains: Theory and Applications*. Ed. by Paula Branco Luís Torgo and Nuno Moniz. Vol. 74. Proceedings of Machine Learning Research. PMLR, 22 Sep 2017, pp. 36–50. URL: <https://proceedings.mlr.press/v74/branco17a.html>.
- [4] Google. *Google Scholar*. <https://scholar.google.com/>. Accessed: 2026-07-21.
- [5] National Library of Medicine. *PubMed*. <https://pubmed.ncbi.nlm.nih.gov/>. Accessed: 2026-07-21.
- [6] Institute of Electrical and Electronics Engineers. *IEEE Xplore Digital Library*. <https://ieeexplore.ieee.org/>. Accessed: 2026-07-21.
- [7] Elsevier. *ScienceDirect*. <https://www.sciencedirect.com/>. Accessed: 2026-07-21.
- [8] Springer Nature. *Springer Nature Link*. <https://link.springer.com/>. Accessed: 2026-07-21.
- [9] Ashenafi Zebene Woldaregay et al. “Data-driven modeling and prediction of blood glucose dynamics: Machine learning applications in type 1 diabetes”. In: *Artificial Intelligence in Medicine* 98 (2019), pp. 109–134. ISSN: 0933-3657. DOI: <https://doi.org/10.1016/j.artmed.2019.07.007>. URL: <https://www.sciencedirect.com/science/article/pii/S0933365717306218>.
- [10] Stella Tsihlaki, Lefteris Koumakis, and Manolis Tsiknakis. “Type 1 Diabetes Hypoglycemia Prediction Algorithms: Systematic Review”. In: *JMIR Diabetes* 7.3 (July 2022), e34699. ISSN: 2371-4379. DOI: 10.2196/34699. URL: <http://www.ncbi.nlm.nih.gov/pubmed/35862181>.

- [11] Heydar Khadem et al. “Artificial Intelligence for Blood Glucose Level Prediction in Type 1 Diabetes: Methods, Evaluation, and Emerging Advances”. In: *Sensors* 26.9 (2026). ISSN: 1424-8220. DOI: 10.3390/s26092675. URL: <https://www.mdpi.com/1424-8220/26/9/2675>.
- [12] William P. T. M. van Doorn et al. “Machine learning-based glucose prediction with use of continuous glucose and physical activity monitoring data: The Maastricht Study”. In: *PLOS ONE* 16.6 (June 2021), pp. 1–17. DOI: 10.1371/journal.pone.0253125. URL: <https://doi.org/10.1371/journal.pone.0253125>.
- [13] William L Clarke et al. “Evaluating Clinical Accuracy of Systems for Self-Monitoring of Blood Glucose”. In: *Diabetes Care* 10.5 (Sept. 1987), pp. 622–628. ISSN: 0149-5992. DOI: 10.2337/diacare.10.5.622. eprint: <https://diabetesjournals.org/care/article-pdf/10/5/622/515265/10-5-622.pdf>. URL: <https://doi.org/10.2337/diacare.10.5.622>.
- [14] Simone Del Favero, Andrea Facchinetti, and Claudio Cobelli. “A Glucose-Specific Metric to Assess Predictors and Identify Models”. In: *IEEE Transactions on Biomedical Engineering* 59.5 (2012), pp. 1281–1290. DOI: 10.1109/TBME.2012.2185234.
- [15] Dave Darpit et al. “A Personalized Federated Learning-Based Glucose Prediction Algorithm for High-Risk Glycemic Excursion Regions in Type 1 Diabetes”. In: *Scientific Reports* 15.1 (Nov. 3, 2025), p. 38376. ISSN: 2045-2322. DOI: 10.1038/s41598-025-22316-4. URL: <https://doi.org/10.1038/s41598-025-22316-4>.
- [16] Nitesh V. Chawla et al. “SMOTE: Synthetic Minority Over-Sampling Technique”. In: *Journal of Artificial Intelligence Research* 16 (2002), pp. 321–357. DOI: 10.1613/jair.953. URL: <https://doi.org/10.1613/jair.953>.
- [17] Hui Han, Wen-Yuan Wang, and Bing-Huan Mao. “Borderline-SMOTE: A New Over-Sampling Method in Imbalanced Data Sets Learning”. In: *Advances in Intelligent Computing*. Ed. by De-Shuang Huang, Xiao-Ping Zhang, and Guang-Bin Huang. Berlin, Heidelberg: Springer Berlin Heidelberg, 2005, pp. 878–887. ISBN: 978-3-540-31902-3.
- [18] Haibo He et al. “ADASYN: Adaptive synthetic sampling approach for imbalanced learning”. In: *2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence)*. 2008, pp. 1322–1328. DOI: 10.1109/IJCNN.2008.4633969.
- [19] Sheng Chen and Haibo He. “SERA: Selectively recursive approach towards nonstationary imbalanced stream data mining”. In: *2009 International Joint Conference on Neural Networks*. 2009, pp. 522–529. DOI: 10.1109/IJCNN.2009.5178874.

- [20] Jiawei Ren et al. “Balanced MSE for Imbalanced Visual Regression”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. June 2022, pp. 7926–7935.
- [21] Luís Camacho, Georgios Douzas, and Fernando Bacao. “Geometric SMOTE for regression”. In: *Expert Systems with Applications* 193 (2022), p. 116387. ISSN: 0957-4174. DOI: <https://doi.org/10.1016/j.eswa.2021.116387>. URL: <https://www.sciencedirect.com/science/article/pii/S095741742101678X>.
- [22] Michael Mayo, Lynne Chepulis, and Ryan G. Paul. “Glycemic-aware metrics and oversampling techniques for predicting blood glucose levels using machine learning”. In: *PLOS ONE* 14.12 (Dec. 2019), pp. 1–19. DOI: 10.1371/journal.pone.0225613. URL: <https://doi.org/10.1371/journal.pone.0225613>.
- [23] Josep Noguer et al. “Generation of Individualized Synthetic Data for Augmentation of the Type 1 Diabetes Data Sets Using Deep Learning Models”. In: *Sensors* 22.13 (2022). ISSN: 1424-8220. DOI: 10.3390/s22134944. URL: <https://www.mdpi.com/1424-8220/22/13/4944>.
- [24] Ciro Rodriguez-Leon et al. “Assessing Bias in the Evaluation of Blood Glucose Prediction Models”. In: *Advances in Computational Intelligence*. Ed. by Ignacio Rojas, Gonzalo Joya, and Andreu Catala. Cham: Springer Nature Switzerland, 2026, pp. 653–663. ISBN: 978-3-032-02725-2.
- [25] The imbalanced-learn developers. *imbalanced-learn Documentation*. <https://imbalanced-learn.org/stable/index.html>. Version 0.14.2, accessed July 18, 2026. 2026.
- [26] The scikit-learn developers. *Common Pitfalls and Recommended Practices*. https://scikit-learn.org/stable/common_pitfalls.html. scikit-learn 1.9.0 documentation, accessed July 18, 2026. 2026.
- [27] Ahmad S. Tarawneh et al. “Stop Oversampling for Class Imbalance Learning: A Review”. In: *IEEE Access* 10 (2022), pp. 47643–47660. DOI: 10.1109/ACCESS.2022.3169512.
- [28] Ibraheem M. Alkhawaldeh, Ibrahim Albalkhi, and Abdulqadir Jeprel Naswhan. “Challenges and limitations of synthetic minority oversampling techniques in machine learning”. In: *World Journal of Methodology* 13.5 (Dec. 2023), pp. 373–378. ISSN: 2222-0682. DOI: 10.5662/wjm.v13.i5.373.
- [29] Wikipedia contributors. *Holm–Bonferroni method* — *Wikipedia, The Free Encyclopedia*. [Online; accessed 19-July-2026]. 2026. URL: https://en.wikipedia.org/w/index.php?title=Holm%E2%80%93Bonferroni_method&oldid=1352765388.

- [30] Glassdoor. *Sueldos para el puesto de Junior Data Scientist en España*. Salary estimates based on salaries anonymously reported to Glassdoor. URL: https://www.glassdoor.es/Sueldos/junior-data-scientist-sueldo-SRCH_K00,21.htm (visited on 07/21/2026).
- [31] Vantage. *g4dn.2xlarge Pricing and Specs*. Amazon EC2 instance with one NVIDIA T4 GPU, 32 GiB of RAM, and an on-demand price starting at \$0.752 per hour. 2026. URL: <https://instances.vantage.sh/aws/ec2/g4dn.2xlarge> (visited on 07/21/2026).

